

University of Strathclyde Glasgow

A Multi-Method Approach to Player Profiling: Integrating Mixed Effects Models, Principal Component Analysis, and Discriminant Analysis in Field Hockey

Amit Prajapati

M.sc Sports Data Analytics

2025 – 2026

Table of Contents

CHAPTER 1: INTRODUCTION 8
1.1 Background 8
1.2 Problem Statement 8
1.3 Research Aims and Questions 9
1.4 Significance of the Study 10
CHAPTER 2: LITERATURE REVIEW 11
2.1 Introduction 11
2.2 GPS-Based Physical Profiling in Team Sports 11
2.2.1 GPS Technology in Field Hockey 11
2.2.2 Positional Differences in Field Hockey 12
2.2.3 Limitations of Traditional Positional Classification 12
2.3 Mixed Effects Models in Sports Science 13
2.3.1 Why Mixed Models Are Necessary 13
2.3.2 Advantages of Mixed Models 13
2.3.3 Applications in Sports 13
2.3.4 Key Measures for Mixed Models 14
2.4 PCA and Clustering for Player Profiling 14
2.4.1 Why Dimensionality Reduction Is Needed 14
2.4.2 PCA Assumptions 14
2.4.3 PCA Applications in Team Sports 14
2.4.4 Clustering for Archetype Identification 15
2.5 LDA and Decision Trees for Position Prediction 15
2.5.1 Linear Discriminant Analysis 15
2.5.2 Decision Trees as a Non-linear Alternative 16
2.5.3 Cross-validation for Small Samples 16

2.6 The Multi-Method Pipeline: Why a Single Approach Is Insufficient 16
2.7 Research Gap and Novelty 17
CHAPTER 3 : METHODOLOGY 19
3.1 Research Design 19
3.2 Participants 19
3.3 Data Collection 19
3.4 Data Processing and Cleaning 20
3.5 Feature Engineering 21
3.6 Statistical Analysis 22
3.6.1 Mixed Effects Models 22
3.6.2 Principal Component Analysis (PCA) 22
3.6.3 Clustering Analysis (K-means) 23
3.6.4 Linear Discriminant Analysis (LDA) 23
3.6.5 Decision Tree Classification 24
CHAPTER 4 : RESULTS 25
4.1 DESCRIPTIVE STATISTICS BY POSITION 25
4.2 DATA DISTRIBUTION CHARACTERISTICS 25
4.3 Mixed Effects Models 27
4.3.1 Intraclass Correlation and Model Comparison 27
4.3.2 Fixed Effects – Match vs Training 28
4.3.3 Fixed Effects – Season Month 29
4.3.4 Random Effects (Player Deviations) 30
4.3.5 Model Fit and Diagnostics 32
4.4 Principal Component Analysis 33
4.4.1 PCA Suitability 33
4.4.2 Variance Explained 33
4.4.3 Loadings and Component Interpretation 35
4.4.4 PCA Biplot and Player Distribution 36
4.4.5 Component Scores Summary 37
4.5 CLUSTERING ANALYSIS 37
4.5.1 Determining Optimal Clusters 37
4.5.2 Cluster Sizes and Composition 38

4.5.3 Player Archetypes 39
4.5.4 Summary of Archetypes 41
4.6 LINEAR DISCRIMINANT ANALYSIS (LDA) 41
4.6.1 LDA Classification Accuracy 41
4.6.2 Confusion Matrix 42
4.6.3 Position Separation 43
4.6.4 Variable Importance 44
4.6.5 Misclassified Players 45
4.6.6 Comparison with Cluster-Based LDA 45
4.7 DECISION TREE ANALYSIS 46
4.7.1 Decision Tree Performance 46
4.7.2 Confusion Matrix and Misclassifications 46
4.7.3 Decision Tree Structure and Rules 48
4.7.4 Variable Importance and Model Comparison 50
CHAPTER 5: DISCUSSION 53
5.1 Summary of Key Findings 53
5.2 PC2 (Explosiveness) as the Primary Position Discriminator 53
5.2.1 Why PC2 Is the Most Substantive Result 53
5.2.2 Why Explosiveness Differentiates Positions 54
5.2.3 Comparison with Previous Research 54
5.3 Alignment and Divergence Between Archetypes and Positions 54
5.3.1 Alignment: Specialists and Strikers 54
5.3.2 Divergence: Balanced Performers Across All Positions 54
5.3.3 What This Reveals About Positional Function 55
5.4 The Elite Performer Cluster: Insightful but Not Generalisable 55
5.5 Methodological Considerations 56
5.5.1 KMO Issue 56
5.5.2 Decision Tree Overfitting 56
5.6 Methodological Strengths and Limitations 56
5.6.1 Strengths 56
5.6.2 Limitations 57
5.7 Recommendations for Future Research 58

CHAPTER 6: CONCLUSION 61
6.1 Summary of Key Findings 61
6.2 Answers to Research Questions 61
6.3 Contribution to Knowledge 62
6.4 Practical Implications 62
6.5 Final Statement 62
REFERENCES 64
APPENDIXES 69
Appendix A 69
Appendix B 70
Appendix C 72
Appendix D 73

Abstract

Background: Hockey is a high-intensity, sporadic sport that requires different physical skills for each position. Traditional positional names (Back, Midfielder, Striker) can mask important within-position variance because they are centred on tactical roles rather than objective physical attributes.

The study's objectives were to: (1) Apply PCA to identify the underlying physical dimensions of hockey performance; (2) Apply clustering to derive data-driven player archetypes; (3) Apply mixed effects models to quantify individual player differences accounting for session context; and (4) Apply LDA and decision trees to assess how well PCA components predict playing position.

Methods: GPS data from 19 elite male field hockey players across 397 sessions (171 matches, 226 training sessions) were analysed. Ten feature-engineered variables (per-minute rates and ratios) were derived. A five-stage analytical pipeline was implemented: Linear Mixed Effects Models (player random intercepts, session type and season month fixed effects), PCA (with varimax rotation), K-means clustering (on PCA components), LDA, and decision tree classification. Assumption testing included KMO, Bartlett's test, ICC, and cross-validation.

Results: ICC=0.270 indicated that 27.0% of the variance in high-intensity running was due to differences between players. PCA reduced the ten variables to three components explaining 84.0% of the variance: PC1 “Workload Volume” (43.3%), PC2 “Explosiveness” (28.7%) and PC3 “Agility” (11.9%). The KMO was 0.443 (Bartlett’s test $p < 0.001$). Three player archetypes were identified: Elite Performer (n=1, PC1=-6.47), Balanced Performers (n=12, mean PC1=-0.14, PC2=0.83) and Specialists (n=6, mean PC1=1.36, PC2=-1.66). LDA predicted location with an accuracy of 73.7% (cross-validation: 57.9%) using PC2 as the primary discriminator (coefficient = -0.657, accounting for 93.3% of between-group variance). The accuracy of the decision tree training decreased from 94.7% to 68.7% under 5-fold cross-validation.

In summary, the multi-method approach revealed three player archetypes that transcend conventional positional constraints and three interpretable physical dimensions. The primary distinction between positions is PC2 (Explosiveness). The

data-driven profiles and traditional locations have a reasonable agreement (73.7%), indicating that positional labels are not perfect representations of physical ability. For position-specific training design, practitioners should use PC2 scores and take individual variations into account instead of positional averages.

Keywords: GPS, Athlete monitoring, Field hockey, player profiling, principal component analysis, mixed effects models, linear discriminant analysis, clustering.

CHAPTER 1: INTRODUCTION

1.1 Background

Hockey is a team sport of intermittent high intensity, where players often sprint, accelerate, decelerate and change direction rapidly during the 60 minute game [1]. Playing position has a major effect on the physical demands. Midfielders cover most distance in total, backs accelerate and decelerate faster and forwards tend to accumulate more sprint distance in general [2], [3]. Tactical role also account for these positional differences. However, players within the same Positional labels such as Back, Midfielder & Striker can produce quite different physical outputs, meaning individual variances are often hidden by positional means [4].

Wearable GPS technology has revolutionised professional sport performance tracking by providing detailed external load measures such as total distance, sprint distance, high-intensity running, accelerations and decelerations [5]. These data move beyond general positional assumptions to data-driven profiling based on actual physical outputs, allowing the identification of underlying physical characteristics and player archetypes that may better reflect individual abilities than traditional positional classifications [6].

1.2 Problem Statement

The main question this study tries to answer is whether data-driven physical archetypes in the top men's hockey leagues align with or deviate from traditional positional duties.

Previous research has examined positional variation with GPS measures, often assigning different physical outputs for forwards, midfielders and backs [3]. Most research, however, assumes that players in the same position have similar physical needs and that positions are stable categories. This assumption might not be valid, as there can be considerable variance within positions due to player-specific characteristics, tactical systems and individual playing styles [7].

Furthermore, to our knowledge, no previous study has integrated clustering, classification, Principal Component Analysis and mixed effects modelling into one analytical pipeline for hockey profiling. Clustering can identify natural player groupings,

PCA can reduce multicollinearity and identify latent physical dimensions [8], mixed models can account for contextual factors (e.g. training vs. match, season progression) [9], [10] and classification techniques can determine if these archetypes match positional roles [11].

1.3 Research Aims and Questions

Research Questions	**Aims**
RQ1: What are the physical characteristics hockey performance is based on?	To Identify the Fundamental Physical Dimensions To discover the latent physical variables that characterise hockey performance using Principal Component Analysis.
RQ2: How do individual players differ physically when considering session context?	Measure individual player variations to measure the differences between individual players, after accounting for session context, using linear mixed effects models.
RQ3: What natural player archetypes appear when players are clustered based on their physical characteristics?	Build Player Archetypes from Data Apply K-means clustering to find natural player groups based on physical features
RQ4: How well do these data driven archetypes align with traditional playing positions?	Compare Archetypes with Traditional Positions To evaluate the relationship between data-driven archetypes and traditional positions, we use Linear Discriminant Analysis.
RQ5: How well do PCA components predict playing position?	Analysing the Prediction Power of PCA Components to evaluate the prediction ability of LDA and Decision Tree for playing position.

1.4 Significance of the Study

This work contributes to applied performance analysis and sports science by integrating multiple models into a single analytical pipeline, providing a novel and comprehensive methodology for player profiling in hockey [8], [9], [11] that can be adapted to other team sports using machine learning and contextual modelling. Empirically, explosiveness, workload volume, and agility are identified as the most salient physical characteristics differentiating roles, and whether data-driven archetypes fit or diverge with established positional structures are evaluated [1], [3]. These results have practical value for talent identification, position-specific training design and player development. They allow coaches to individualise training when explosiveness is a strong defining characteristic of a role, or to avoid uniform prescription when significant within-position diversity exists.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Player profiling in team sports is increasingly based on objective data through wearable technologies and advanced statistical modelling. Mixed effects models, PCA, clustering and classification techniques allow better understanding of positional demands and individual differences. GPS-based monitoring in hockey provides in-depth information on external load [5]. The 4 main topics covered in this chapter are: (1) hockey positional differences and GPS based physical profile, (2) mixed effects models for repeated measures sports data, (3) player archetype identification using PCA and clustering and (4) position prediction using LDA and decision trees. The conclusion of the chapter describes the research gap. Namely, the lack of integrated multi-method pipelines with mixed models, PCA, clustering and classification in top hockey.

2.2 GPS-Based Physical Profiling in Team Sports

2.2.1 GPS Technology in Field Hockey

Global Positioning System (GPS) technology is currently used in elite team sports to quantify external stress, including total distance, high-speed running, sprint distance, accelerations and decelerations [5], [12]. Systematic evaluations have reported the extensive usage of GPS and related microtechnology sensors in football, rugby and hockey to track training and game demands [5].

According to wearable technology assessments, GPS units are the most popular devices used to collect activity profiles, notably in hockey. However, sampling rates and device models vary between research, making direct comparisons problematic [13]. Peak speed and extremely short, high intensity movements are more likely to be subject to measurement error, however validation work in team sports suggests that distance-based measures are often trustworthy [14]. These results provide an explanation for the omission of very brief workouts, or partial sessions, when comparing training and match needs, and support the inclusion of session level GPS measures.

2.2.2 Positional Differences in Field Hockey

GPS has been employed in several studies to investigate positional variability in hockey Forwards in elite women's hockey have been shown to spend more time at high intensities than defenders and midfielders [3] due to their involvement in pressing and counterattacking. There is large variation in total distance, high-speed running and sprinting outputs of strikers, midfielders and defenders in international men's hockey, with midfielders tending to cover the greatest total distance and forwards sprinting the most [1].

Fixture congestion also impacts positional burden. Research on the compressed schedules in hockey has shown that, although midfielders keep a high total distance throughout games, forwards run a higher sprint distance and high intensity running and sprint measures are very sensitive to short recovery times [15]. These results make it possible to predict season progression and match vs training context during GPS data analysis.

2.2.3 Limitations of Traditional Positional Classification

Most positional profile studies treat positions as fixed, discrete categories and assume that all players in a role have similar physical requirements. However, individual traits, playing styles and tactical systems [4], [6] can cause great variation within positions. For example, a striker playing in a possession-based system may perform more short accelerations and decelerations, while a striker playing in a counterattacking team may have a sprint-dominant profile.

This limitation motivates data-driven methods that find archetypes not on the basis of predefined labels, but on the physical profiles. In this study, we present a way to determine and evaluate player archetypes in top hockey. The method combines mixed models, PCA, clustering and classification.

2.3 Mixed Effects Models in Sports Science

2.3.1 Why Mixed Models Are Necessary

Sports performance data are naturally hierarchical: players are nested within teams or tournaments; players are nested within many sessions. This structure violates the independence assumption of both conventional repeated measures ANOVA and ordinary least squares regression [9]. This layering can be explicitly modelled with mixed effects models (also known as multilevel models) by including random effects for individuals, allowing repeated measures to be properly modelled [9], [10].

2.3.2 Advantages of Mixed Models

For sports data, linear mixed models have the following benefits:
By allowing each player to have their own baseline performance level, random intercepts capture consistent changes across players [9].
Players can react differently to contextual elements like season month or session type (match vs. training) due to random slopes [9].
Unbalanced data handling prevents listwise deletion by enabling the inclusion of players who have missed sessions because of rotation or injury [9].

Because it offers broad specification of random structures and effective estimates, the R lme4 package has become a common tool for fitting mixed models in practical sports research [9].

2.3.3 Applications in Sports

Mixed models have been used to quantify contextual factors such as match location, opposition strength and fixture congestion and to distinguish between athlete variability and between athlete differences in rugby, football and other team sports [4], [5], [9]. Similarly, in golf, mixed models with player-specific random intercepts have better performance trajectory prediction than fixed effects only models [9]. These applications demonstrate the suitability of mixed models to repeated measures GPS data in hockey.

2.3.4 Key Measures for Mixed Models

The intraclass correlation coefficient (ICC) is the proportion of total variability that is due to player differences . The guidelines [16], [17] indicate that ICC values less than 0.05 indicate little clustering, 0.05–0.20 some, 0.20–0.50 moderate and greater than 0.50 a lot of clustering.

Authors defined conditional and marginal R^2 for mixed models, with the conditional R^2 representing the variance explained by both fixed and random factors, and the marginal R^2representing the variance explained by fixed effects alone. After controlling for session context, the difference between these values is the extra variance explained by individual level random effects when profiling players[18].

2.4 PCA and Clustering for Player Profiling

2.4.1 Why Dimensionality Reduction Is Needed

Many GPS derived performance measures are highly correlated (e.g. distance, high intensity running, sprint distance, accelerations, decelerations) and have been used as an indicator of performance. This multicollinearity can lead to instability of regression models [8], [17]. This is achieved by PCA which reduces correlated variables to a smaller number of uncorrelated components that explain most of the variation in the data [8], [19]. PCA helps identifying latent physical characteristics like agility, explosiveness and workload volume for player profiling.

2.4.2 PCA Assumptions

Standard diagnostics of suitability for PCA include Bartlett's test of sphericity and the Kaiser Meyer Olkin (KMO) measure of sample adequacy. The significant Bartlett's test indicates that the PCA is sufficient and the correlation matrix is not an identity matrix. The KMO value above 0.6 is generally acceptable [19].

2.4.3 PCA Applications in Team Sports

PCA has been used in hockey, rugby and football to identify key performance indicators and to reduce dimensionality. PCA has accounted for a high proportion of the variance of physical testing data in professional soccer and has identified components

suggesting muscular strength, aerobic ability and body composition [20]. Multivariate profiling in rugby has been investigated using PCA and identified factors related to technical engagement, running intensity and collision load [7].

In injury surveillance, PCA and clustering have been used to reduce biomechanical variables to a reasonable number of components before the discovery of risk-related clusters in elite female football players [21], [22]. These examples are similar to the use of PCA in the present study on player-adjusted GPS measures before clustering.

2.4.4 Clustering for Archetype Identification

Clustering discovers the natural groupings in the reduced component space and PCA reduces the dimensionality. K means clustering, which assumes roughly spherical clusters in Euclidean space, is often applied to continuous data [11]. In sports, clustering has been used to study injury risk, physical attributes and tactical tendencies.

For example, cluster analysis has been used to identify shoulder injury risk groups in overhead athletes, and has shown discrete clusters with different injury rates [23]. Clustering has been used in invasion sports to identify tactical archetypes and playing styles that are not strictly according to traditional positions [6]. Robustness is increased by taking into account several factors for the selection of the optimal number of clusters, such as gap statistics, silhouette scores, and the elbow technique [11].

These methods can be combined into a logical pipeline for profiling and classification [7] and have been used sequentially in multivariate data exploration including PCA, clustering and discriminant analysis.

2.5 LDA and Decision Trees for Position Prediction

2.5.1 Linear Discriminant Analysis

Linear discriminant analysis (LDA) is a supervised classification method that finds linear combinations of predictors that best separate pre-specified groups. It assumes equality of covariance matrices between groups and multivariate normality [11]. In sports, the LDA has been used to classify playing positions based on physical and technical parameters.

Woods used LDA to indicators of technical skill in elite junior Australian football and successfully classified players to roles with a ~70% accuracy, with the ability to distinguish forwards from defenders, but not overlapping roles. The results illustrate the advantages and disadvantages of linear categorisation in cases of similar role demands [7], [24].

2.5.2 Decision Trees as a Non-linear Alternative

Decision trees are nonlinear classification models that recursively partition the predictor space into regions defined by simple rules [11], [25]. They are robust to outliers, do not require distributional assumptions and are highly interpretable decision rules, which is beneficial for coaches and practitioners.

LDA and decision trees have been compared and it has been observed that both can achieve similar accuracy in sporting contexts, however decision trees tend to provide more practical rules [7]. In this study, decision trees are used combined with LDA to verify if the correlations between PCA components and playing position are linear or non-linear.

2.5.3 Cross-validation for Small Samples

Therefore, cross-validation is a crucial step in assessing generalisability, as single team case studies are often based on very small sample sets. When the number of observations is less than 50, K fold cross validation is recommended where it balances the bias and variance of performance estimates [11], [26]. This study compares the training and validation accuracy of decision trees using cross validation to identify possible overfitting.

2.6 The Multi-Method Pipeline: Why a Single Approach Is Insufficient

Each methodological element addresses a separate constraint that cannot be resolved by one strategy:

Method	Addressess	Why Needed
Mixed Effect Models	For session context (training vs. match, season development)	ensure that differences in exposure do not bias player comparisons (Bates et al., 2015; Bradley & Noakes, 2013)[4], [9].
PCA	Multicollinearity among GPS metrics	PCA detects latent physical dimensions and addresses it(Kaiser, 1974; Jolliffe, 2002)[8], [19].
Clustering	Variance within positions	identify data-driven archetypes that may deviate from traditional roles (Rein & Memmert, 2016)[6].
LDA & Decision Tree	The predictive power of PCA components	To confirm whether the data-driven archetypes correspond to conventional positional views (Woods et al., 2018)[7].

2.7 Research Gap and Novelty

A clear gap in the literature is that there is no positional profiling in elite hockey that incorporates PCA, mixed effects modelling, clustering and classification in a single framework. In particular:

• GPS studies in hockey have reported positional differences, but mixed models are rarely used to control for session type or season progression [1], [3].
• Mixed models are often used in other sports, but hockey player profiling has not been consistently applied [9].

• PCA and clustering have been used separately on rugby and soccer, but not integrated as a pipeline for field hockey [7], [20].

The present study addresses this gap by:

1	Deviations from player level are estimated in mixed models controlling for session type and season month.
2	Derivation of basic physical dimensions (e.g. workload volume, explosiveness, agility) through PCA on player-removed GPS data.
3	Player archetypes are derived through k means clustering of PCA scores.
4	Validate archetypes with traditional perspectives, assess forecast accuracy with LDA and decision trees.

This multi-method pipeline contributes to hockey research methodologically, as it demonstrates how complementary methodologies can be coupled to provide richer, coach-relevant insights than any one method alone.

CHAPTER 3 : METHODOLOGY

3.1 Research Design

In the current study, a quantitative, retrospective, and observational approach with GPS data collected for Uddingston elite men's field hockey team for one competitive season (September 2025 - March 2026) was implemented. To perform the profiling analysis, an analytical pipeline consisting of multiple analytical techniques was designed, including mixed effects models, principal component analysis (PCA), clustering, LDA, and decision tree algorithms. The use of such a variety of methods is justified by the need to address distinct aspects of player profiling: Mixed models take into account repeated measures [9], PCA is used to reduce the number of variables [27], clustering finds groups [28], and classification validates the clusters against the positional categories of players [25], [29].

3.2 Participants

A total of 20 elite men's field hockey players of a Uddingston Hockey club team's were involved in the current study. All participants were categorized according to their playing position: Backs (n=2), Midfielders (n=8), and Strikers (n=9). Missing data on positions was observed in one participant, who was subsequently excluded from further analyses. This sample size is quite small; however, it is relatively common in single team cases when studying elite athletes since recruiting a large cohort of players is not feasible due to practical constraints [5], [30].

3.3 Data Collection

The GPS data were gathered through wearable devices (PlayerData) that participants wore during all training sessions and competitive matches within one competitive season (September 2025 - March 2026). The devices had a sampling rate of 10 Hz and measured the following variables: total distance (m), distance covered during high-speed running (m; >5.5 m/s), sprint distance (m; >7.0 m/s), accelerations and decelerations (counts), session load (arbitrary units), top speed (m/s), average speed (m/s), and percentage of maximum speed. Sessions lasting less than 60 minutes were

excluded as the comparison of such short drills to matches could lead to bias caused by differences in the durations [14]. GPS units with 10 Hz sampling frequency have been shown to provide acceptable validity for team-sport movement analysis [14], [31].

3.4 Data Processing and Cleaning

Several consecutive data processing and cleaning operations were carried out on the data. First, rows where important performance metrics had missing values were located, and 12 rows with missingness in all numeric data were deleted since they reflected periods of non-wear or malfunction of GPS devices. Second, percentages of maximum speed above 120% were corrected to the median value of existing data to handle outliers that are physiologically unrealistic [17]. Third, duplicates were removed through unique identification of each row by player ID, date, time, and session type. Fourth, the data were restricted to whole sessions lasting at least 60 minutes to exclude quarter, half, and drill data which would skew the comparisons due to differences in duration. As a result, the data set size decreased from 955 observations down to 397 session-level observations corresponding to 20 players, with 171 match sessions and 226 training sessions in total. Missing values in numeric columns were imputed by their respective column mean, and no data was excluded because of their being outliers since this way, the full variation of performance typical to elite sports is maintained [32].

3.5 Feature Engineering

Table 3.1: Feature Engineering Variables

Feature-Engineered Variables

Feature	Formula	Purpose
hir_per_min	hir_m / duration_mins	High-intensity running rate
sprint_per_min	sprint_m / duration_mins	Sprinting frequency
accels_per_min	accels / duration_mins	Acceleration rate
decels_per_min	decels / duration_mins	Deceleration rate
distance_per_min	distance_m / duration_mins	Overall pace
intensity_ratio	hir_m / distance_m	Proportion of high-intensity work
sprint_ratio	sprint_m / distance_m	Proportion of sprinting
accel_decel_ratio	accels / decels	Balance of accelerations vs decelerations
work_rate	session_load / duration_mins	Internal load per minute
speed_endurance	top_speed / avg_speed	Ability to maintain speed

In order to correct for the session duration issue and consider rate-based performance metrics, 10 feature engineered variables were created based on raw GPS data. The creation of this feature set was needed because accumulated metrics (total distance, total HIR, etc.) do not provide reliable information since, for example, a 90-minute training session is expected to yield higher raw values than 60 minutes of a match despite the match being more intense. Therefore, rate-based features were considered – high-intensity running per minute, sprint distance per minute, accelerations per minute, decelerations per minute, and distance per minute. Rate-based metrics are widely recommended in team-sport GPS analysis to normalise for session duration [31], [32]. In addition, ratio features were considered – intensity ratio (HIR/Total Distance), sprint ratio (sprints/Total Distance), and accel-decel ratio (accelerations/decelerations). The remaining two were work rate (session load/duration) and speed endurance (top speed/average speed). Infinite numbers caused by division by zero were replaced by NA and imputed afterwards using column mean.

3.6 Statistical Analysis

3.6.1 Mixed Effects Models

Linear mixed effects models were used to model the data for two reasons: First, we have a structure of repeated measures (multiple measurements per player); thus, it would violate the independence assumption of ordinary least squares regression if used. In addition, mixed effects models allow for modeling player-level heterogeneity through adding random intercepts for players, which allow different players to have different baselines. A basic specification would be: metric ~ session_type + season_month + (1 | player_id) with session type (match vs training) and season month (1 to 7 representing September to March) being fixed effects, whereas the player ID is a random effect. Mixed-effects modelling is widely used in sports performance research because it accounts for repeated observations and individual variability [9].

A fixed session effect was included to compare match and training demands, and season month was included to capture the fitness–fatigue curve during the season. Random slopes for session type could be added using (1 + session_type | player_id), with model choice based on AIC and likelihood ratio tests.

HIR_per_min was selected for ICC calculation as high-intensity running is a key performance indicator in elite hockey, relevant across all player positions and one of the most sensitive GPS-derived metrics for detecting between-player variability. It is therefore a suitable representative variable to describe the clustering degree in the dataset. The intraclass correlation coefficient estimated the extent to which players differed in their performance metrics (Shrout & Fleiss, 1979)[33]. R^2 marginal and R^2 conditional were used to quantify variance explained by fixed and random effects [18]. Given the warning of a singular fit for Model 2, indicating overfitting of the random effects structure, the final model was Model 1 (random intercepts only), which had more stable estimates and better interpretability.

3.6.2 Principal Component Analysis (PCA)

The PCA was applied to the ten feature engineered variables with scaling to ensure that all the variables were contributing equally (mean = 0, standard deviation = 1). PCA was

selected due to the high correlation between the ten feature variables as indicated by the correlation matrix, causing multicollinearity in regression analysis and making it difficult to interpret the results [27]. In this case, PCA plays an important role in the player profiling as it helps identify the hidden dimensions behind performance like "workload volume" and "explosiveness" [22]. The number of components was obtained using the Kaiser rule and the inflection point in scree plot, while loadings above 0.5 contribute significantly to the interpretation of the components. PCA component scores of the first three components were computed for all players, and these were then used for clustering analysis since it simplifies the clustering procedure to only three dimensions from the ten inputs.

3.6.3 Clustering Analysis (K-means)

The k-means algorithm was used on the first three PCA components, randomly starting the procedure 25 times in order to prevent convergence into a local optimum. This approach was chosen because k-means clustering does not require pre-defined groups and can be viewed as an unsupervised method, answering the question, "what are the natural player archetypes?" [28]. Optimal number of clusters was estimated using three methods: the elbow criterion (within-cluster sum of squares minimum), the silhouette criterion (average silhouette score maximum) [34], and gap statistic [35]. The three-cluster solution was picked according to the criteria. Stability of the clusters was calculated and further interpretation of the obtained player archetypes was done by means of their mean PCA component scores [23].

3.6.4 Linear Discriminant Analysis (LDA)

The choice of LDA was justified for two reasons. Firstly, LDA is a supervised classification method, requiring known groups in the data, which in our case are the playing positions [24], [29]. Secondly, this method assumes normal distribution of the predictors and equality of their covariance matrices between groups, which is reasonably satisfied in this dataset. Specification of the linear discriminant model was the following: position_simple ~ PC1 + PC2 + PC3. Accuracy of the classification was estimated via confusion matrices, while generalisability of the obtained results was estimated through leave-one-out cross-validation, a standard approach for small

datasets [36]. Proportion of trace was calculated in order to measure the amount of variation that each discriminant function explains.

3.6.5 Decision Tree Classification

A decision tree classifier was used to examine whether the relationship between the PCA components and playing position was linear or non-linear. This method was selected as part of an exploratory analytical framework, as no previous field-hockey study has implemented a multi-method pipeline combining mixed models, PCA, clustering, LDA and decision trees for player profiling.

Decision trees were trained with the following hyperparameters: minimum observations to split = 2, minimum observations in a terminal node = 1, complexity parameter = 0.01, and maximum tree depth = 3. Given the small sample size, these settings were chosen to balance exploratory flexibility with control over model complexity. The permissive values for minsplit and minbucket allowed the tree to consider all possible splits in the data, while the complexity parameter (cp = 0.01) and maximum depth (depth = 3) restricted unnecessary branching. However, the permissive splitting rules still contributed to overfitting, as reflected in the drop from training accuracy (94.7%) to cross-validation accuracy (68.7%).

The model predicted playing position from the three PCA components using the formula: position_simple ~ PC1 + PC2 + PC3. Variable importance was calculated based on the reduction in impurity at each split. Five-fold cross-validation was conducted to evaluate potential overfitting, comparing training accuracy against cross-validated accuracy. A large discrepancy between training and cross-validation accuracy indicates that the model has memorised the training data rather than learning generalisable patterns, which is a particular risk with small sample sizes [36].

CHAPTER 4 : RESULTS

4.1 DESCRIPTIVE STATISTICS BY POSITION

Table 4.1: DESCRIPTIVE STATISTICS BY POSITION

Descriptive Statistics by Position

Position	n	Distance (m)	HIR (m)	Sprints (m)	Accels (n)	Decels (n)	Top Speed (m/s)
Back	57	6367 ± 1495	272 ± 253	80	56	59	7.2
Midfield	198	6317 ± 1443	296 ± 155	91	46	49	7.0
Striker	140	5945 ± 1558	383 ± 212	104	48	52	6.8

Table 4.1 summarises the positional differences in physical outputs across 397 sessions. In general the backs produced the greatest volume of work, the midfielders a moderate but balanced output and the strikers the greatest high intensity and sprint efforts. These patterns suggest significant positional differences in elite men's hockey, consistent with previous literature [2].

4.2 DATA DISTRIBUTION CHARACTERISTICS

Skewness and kurtosis values (Appendix A, Table A1) indicate that distance and session load were approximately symmetric, while HIR and sprint distance showed moderate positive skew typical of elite performance distributions. Top speed displayed strong negative skew, and several variables, including HIR, top speed and intensity ratio, were markedly leptokurtic. Work rate was closest to a normal distribution.

The identification of outliers was conducted by the interquartile range (IQR) method (see Appendix A, Table A2) , and outliers were most common in duration, with additional outliers observed in average speed, HIR and sprint distance. All outliers were retained to preserve naturally occurring variability in elite sport datasets [14].

FIGURE 4.1: Distribution of Key Performance Metrics

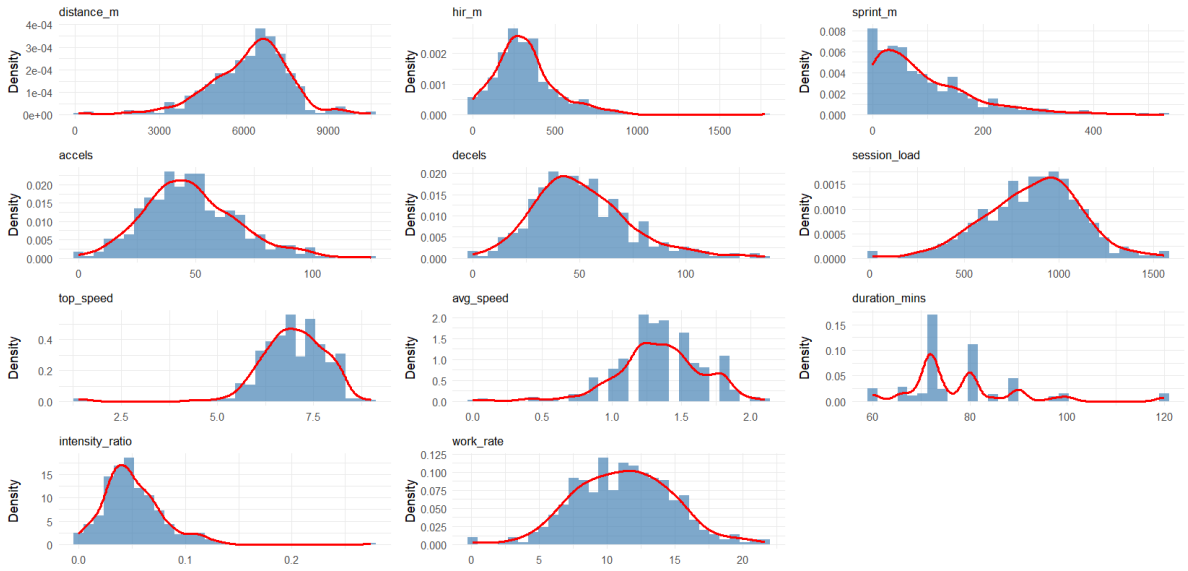


Figure 4.1 shows the distribution of key performance metrics across all 397 sessions. HIR, sprint distance and intensity ratio display positively skewed distributions, maximum velocity is negatively skewed, and work rate is approximately normal.

FIGURE 4.2: Positional Differences in Performance Metrics

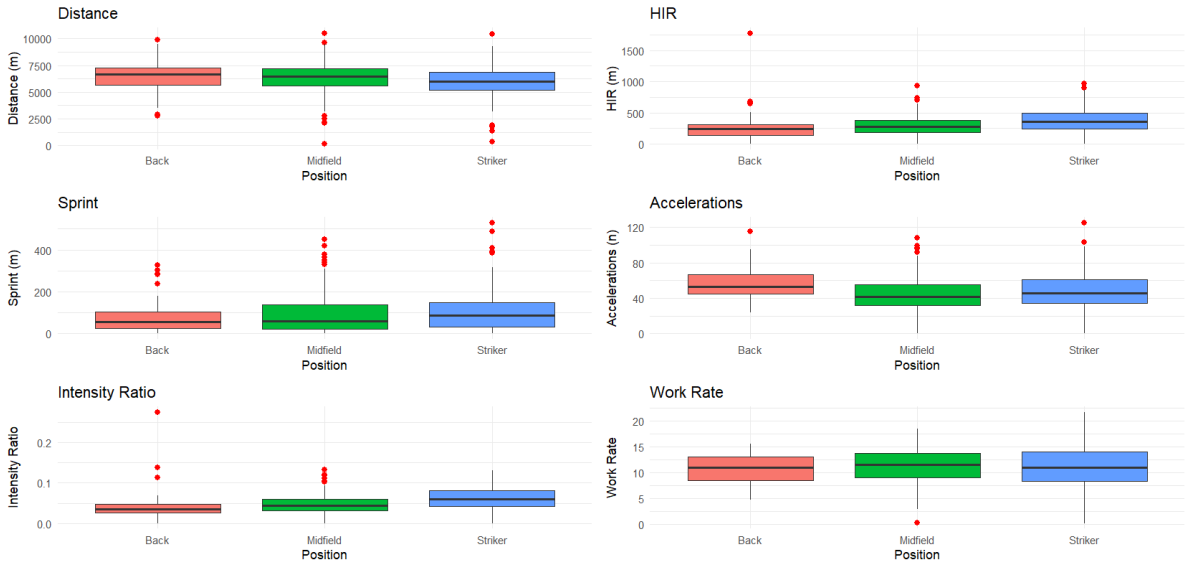


Figure 4.2 illustrates positional differences in performance metrics. Strikers show higher HIR and sprint outputs, midfielders demonstrate stronger work-rate measures, and backs record more accelerations.

4.3 Mixed Effects Models

4.3.1 Intraclass Correlation and Model Comparison

To measure the amount of variance explained by inter-player differences, an intra-class correlation coefficient (ICC) was derived using a null model of hir_per_min, which contained a random intercept term for player only. The resulting ICC value of 0.270 indicated that 27.0% of the total variance in hir_per_min could be attributed to inter-player differences, and the other 73.0% was accounted for by intra-player variance (Table 4.2). The obtained ICC value makes it justified to use the mixed effects models as there is considerable variance explained by players' clusters [9].

TABLE 4.2: Intraclass Correlation and Variance Components

Parameter	Variance	Standard Deviation	Percentage
Player (Intercept)	1.874	1.369	27.0%
Residual	5.063	2.250	73.0%
ICC	0.270	-	-

Note: ICC = 0.270 indicates moderate player-level clustering [16].

Three models were compared based on likelihood ratio test statistics and AICs. The first one, called null model (m0), consisted of only random intercept. Model 1 (m1) had additionally fixed predictors session type (match vs training) and month of season. Model 2 (m2) had also a random effect of session type, meaning that players could differ in individual reaction to either type of sessions. As presented in Table 4.3 below, each consecutive model had better performance than previous model – m1 fitted better than m0 ($\chi^2 = 46.02$, df = 2, $p < 0.001$) and m2 fitted better than m1 ($\chi^2 = 26.77$, df = 2, $p < 0.001$). AICs supported the same hierarchy of models (m0 = 1805.68, m1 = 1763.67, m2 = 1740.89). Unfortunately, m2 provided a warning about singular fit because of model overfitting caused by too complex random effects.

TABLE 4.3: Model Comparison

Model	Description	AIC	LogLik	χ^2	df	p-value
m0	Null (random intercept only)	1805.68	-899.84	-	-	-
m1	Fixed effects (session type + season month)	1763.67	-876.83	46.02	2	<0.001 ***
m2	Random slope (session type | player)	1740.89	-863.45	26.77	2	<0.001 ***

4.3.2 Fixed Effects – Match vs Training

The high-intensity running rate (hir_per_min) in matches was significantly higher than that in training sessions, where the reduction was 1.52 units (95% CI [-1.96, -1.09], $p < 0.001$). Four out of the total ten metrics (Figure 4.3) showed a significant difference between the match and training scenarios, with the values being significantly higher in the former: distance_per_min ($\beta = 14.67$, $p < 0.001$), intensity_ratio ($\beta = 0.0125$, $p < 0.001$), and work_rate ($\beta = 3.47$, $p < 0.001$). On the other hand, training demonstrated higher scores for the following parameters: accels_per_min ($\beta = -0.088$, $p < 0.001$), accel_decel_ratio ($\beta = -0.212$, $p < 0.001$), and speed_endurance ($\beta = -0.853$, $p < 0.001$). For the remaining metrics, sprint_per_min, decels_per_min, and sprint_ratio did not show any statistical significance between the two conditions (all $p > 0.05$). Thus differences between match and training metrics are shown in Figure 4.3. These patterns are explored further in Chapter 5.

FIGURE 4.3: Match vs Training Effect on Performance Metrics

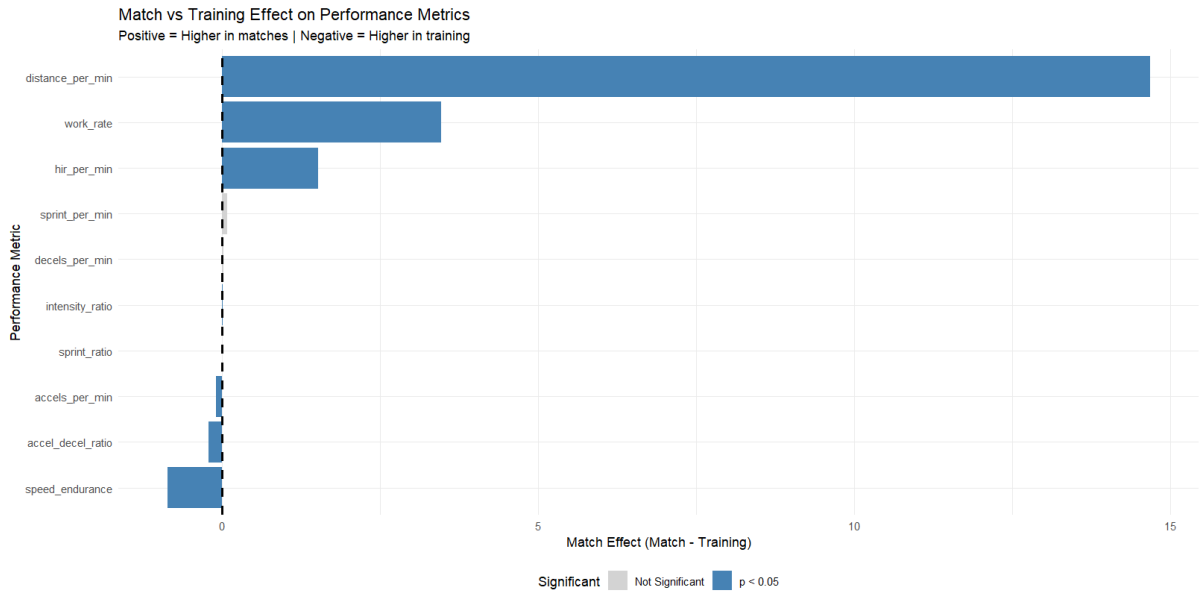


Figure 4.3 shows the match–training effects for all performance indicators. hir_per_min, distance_per_min, intensity_ratio and work_rate were significantly higher during matches, while accels_per_min, accel_decel_ratio and speed_endurance were higher during training.

4.3.3 Fixed Effects – Season Month

The seasonal analysis (Figure 4.4) was performed by examining how performance metrics changed throughout the seven-month competitive season (September 2025-March 2026). Two measures had a statistically significant increase: sprint ratio (β = 0.0010/mo; 95% CI [0.0002, 0.0018]; p = 0.011) and speed endurance (β = 0.164/mo; 95% CI [0.054, 0.273]; p = 0.003), which shows that there was an increase in relative percentage sprint and better maintenance of higher speeds by players in the latter part of the season. In contrast, there were significant decreases in accels_per_min (β = -0.016/mo; 95% CI [-0.025, -0.006]; p = 0.002) and decels_per_min (β = -0.013/mo; 95% CI [-0.026, -0.001]; p = 0.039), this reflects that fatigue accumulation or tactical adjustments in COD demands. Full seasonal results are presented in Appendix B (Table B2).

FIGURE 4.4: Seasonal Improvement/Decline

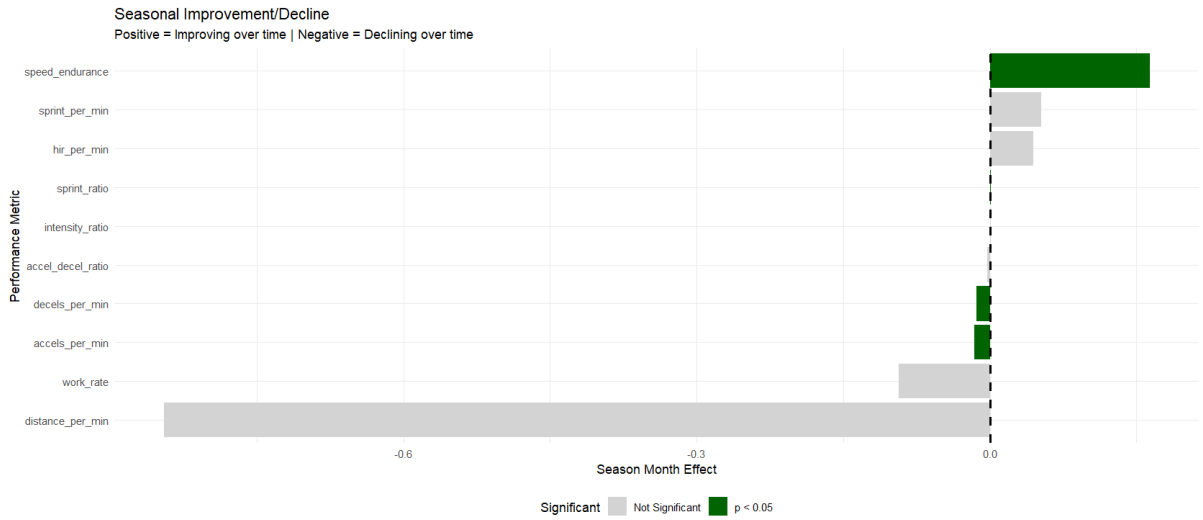


Figure 4.4 shows the seasonal effects on performance variables. Sprint_ratio and speed_endurance improved significantly over the season, while accels_per_min and decels_per_min showed significant declines.

4.3.4 Random Effects (Player Deviations)

The values of the random intercepts obtained from the final model signify the amount of deviation from the mean of high-intensity running by each player once the effect of session type and month is accounted for (Figure 4.5). Angus Millar had the highest deviation (+3.30), indicating a higher adjusted high-intensity rate compared to the team mean. Players with above-average deviation also include Max (+1.34) and Glenn Turner (+1.17). On the other hand, there are players who showed below-average deviation such as Andrew Docherty (-0.013), Calum Darroch (-0.011), and Louis Gardiner (-0.032), although their deviations were relatively small. The range of random effects is shown in Figure 4.5.

FIGURE 4.5: Player-Adjusted High-Intensity Running Rate

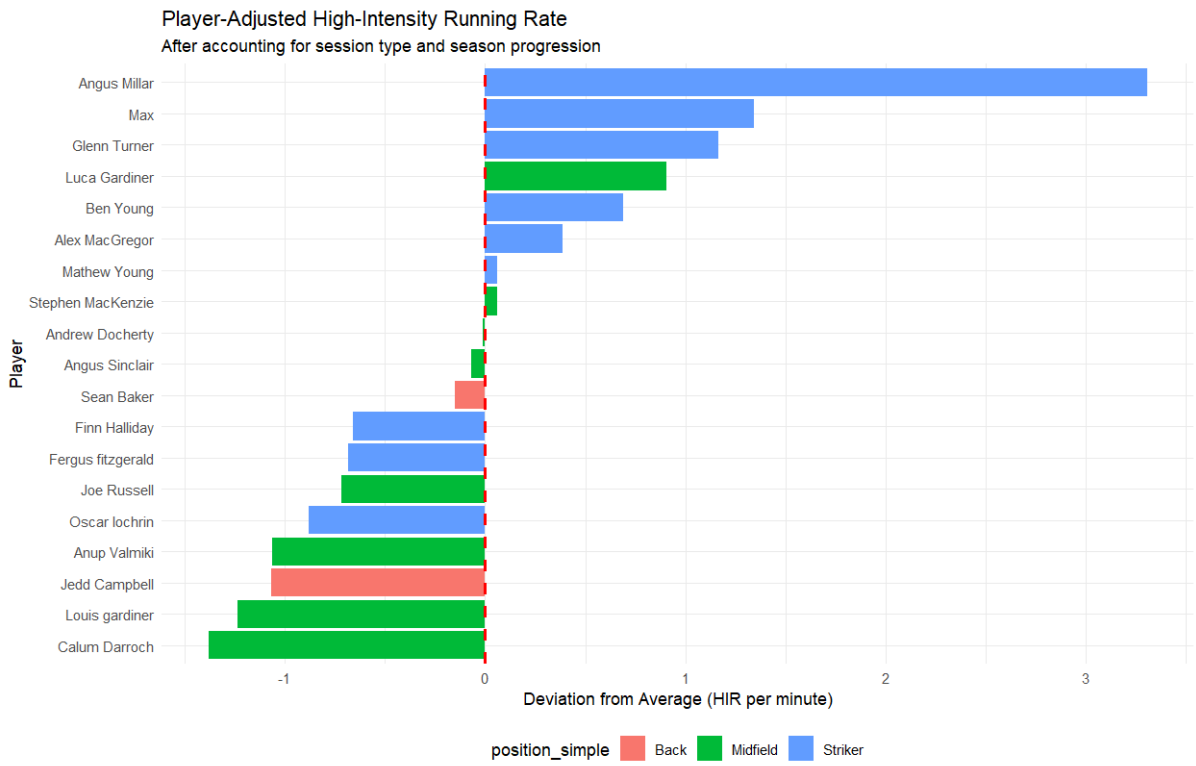


Figure 4.5 shows player deviations from the population average for high-intensity running rate (hir_per_min), adjusted for session type and season month. Angus Millar had the highest positive deviation (+3.30), while Calum Darroch showed the lowest (–1.60).

FIGURE 4.6: Player Performance Deviations Heatmap

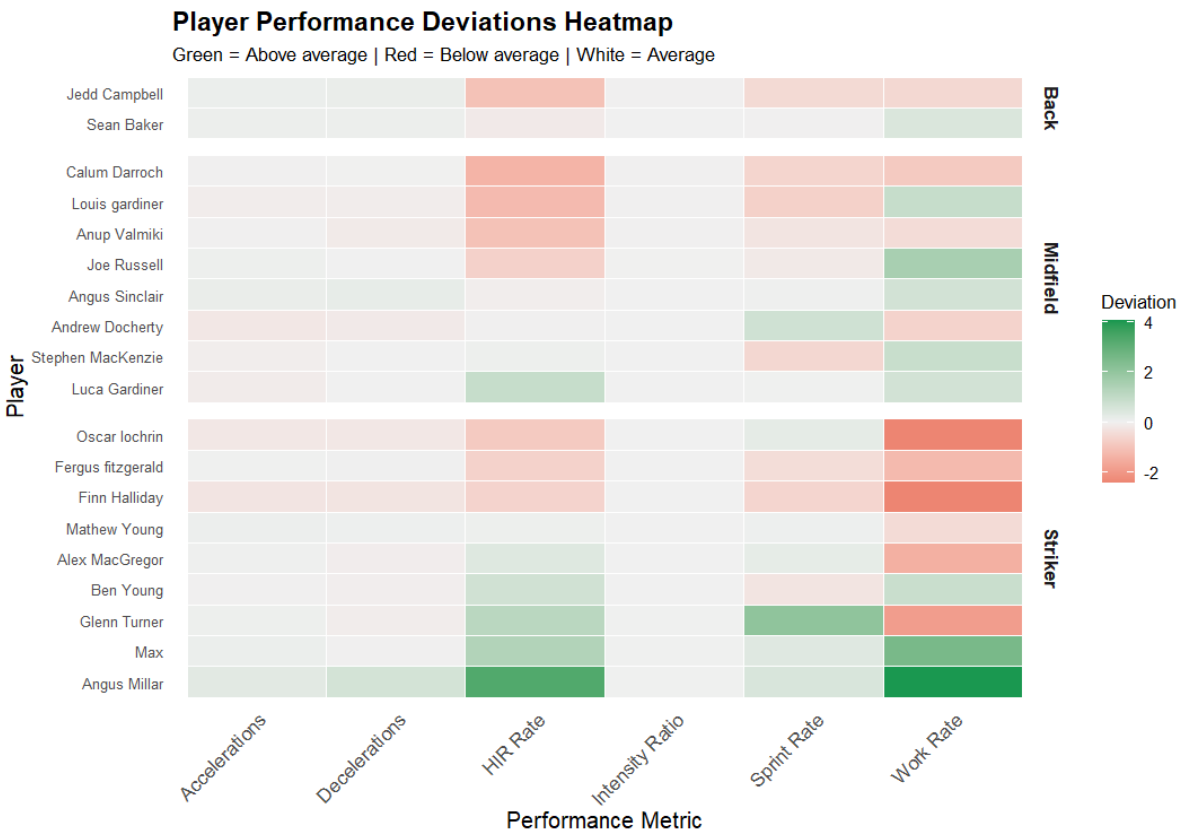


Figure 4.6 provides a deviation map for each player based on the six main KPIs, categorizing them according to playing positions. As seen from the figure, players like Angus Millar (Striker) performed better than their counterparts in almost all KPIs, while other players in positions such as Midfielder and Backs had both strengths and weaknesses specific to their positions.

4.3.5 Model Fit and Diagnostics

The final model accounted for 8.6% of variance using fixed effects alone (marginal R^2) and 32.0% using fixed and random effects together (conditional R^2). It is evident from the high difference between the two R^2 values (23.4 percentage points) that there was a significant contribution from player-level random effects on the variation that was independent of session type and season month [18].

However, residual normality assumption of the linear mixed model was violated according to the Shapiro-Wilk test ($p < 0.001$). Such a situation is typical with sports performance data due to skewness caused by elites in the upper tail of the distribution. There was no heteroscedasticity ($p = 0.055$) nor singularity in the model. Although mixed

models tolerate normality assumption violation when the sample size is more than 30 [10], non-normal residuals remain a weakness of the study mentioned in Chapter 5.

4.4 Principal Component Analysis

4.4.1 PCA Suitability

Before PCA, it was first necessary to determine whether the data could be reduced in dimensions. Two tests were conducted for this purpose: the Kaiser-Meyer-Olkin (KMO) test and the Bartlett's test of sphericity. For the KMO test, the KMO value obtained was 0.443, which is less than the desirable value of 0.60 and suggests that the correlational pattern may not be optimal for PCA [19]. Nevertheless, this problem is taken into account in Chapter 5. On the other hand, Bartlett's test showed that the correlation matrix was far from being an identity matrix, $\chi^2(45) = 246.8$, $p < 1.19e\text{-}29$. Moreover, there were several variable pairs with high correlation between them ($|r| > 0.7$), such as hir_per_min_adj and intensity_ratio_adj ($r = 0.924$) as well as sprint_per_min_adj and sprint_ratio_adj ($r = 0.975$). Despite the modest KMO value, PCA was retained because Bartlett's test confirmed sufficient shared variance and several variables demonstrated very strong correlations ($|r| > 0.7$). Given the exploratory nature of this study, PCA was considered appropriate, although the results should be interpreted with caution.

4.4.2 Variance Explained

Principal component analysis was conducted using the ten player-adjusted performance measures in order to determine the latent factors that affect performance in field hockey. This process helped reduce the ten correlated variables to three principal components, each with an eigenvalue greater than 1.0 and accounting for a total of 84.0% of the total variability (Table 4.4). The percentage of variance explained by each of the three principal components is as follows: PC1 explains 43.3%, PC2 accounts for 28.7%, and PC3 explains 11.9%. The scree plot depicted in Figure 4.7 shows that the inflection point lies at three components.

TABLE 4.4: Variance Explained by PCA Components

Variance Explained by PCA Components

	Component	Variance	Cumulative
PC1	PC1	43.34	43.34
PC2	PC2	28.73	72.07
PC3	PC3	11.89	83.96
PC4	PC4	8.63	92.58
PC5	PC5	4.26	96.84
PC6	PC6	2.37	99.21
PC7	PC7	0.46	99.67
PC8	PC8	0.24	99.90
PC9	PC9	0.08	99.98
PC10	PC10	0.02	100.00

Note: Only components with eigenvalue > 1.0 (PC1–PC3) were retained for interpretation

FIGURE 4.7: Scree Plot

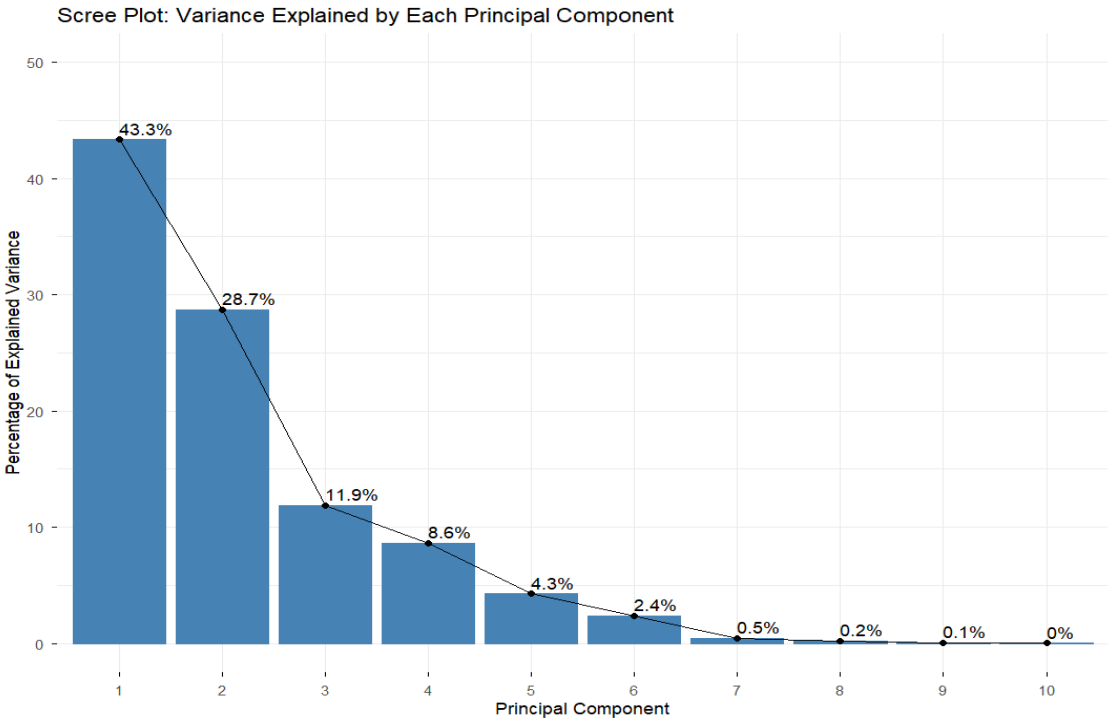


Figure 4.7 shows the variance represented by each principal component. An inflection point occurs at the third component, thus retaining principal components one, two, and three that account for 84.0% of the variance.

4.4.3 Loadings and Component Interpretation

Table 4.5 shows the loadings for the first three principal components. Interpretation was based on strong loadings (|0.5| threshold). PC1 ("Workload Volume", 43.3% variance) represented overall physical output, with high contributions from distance, work rate, high-intensity running and deceleration-related metrics. PC2 ("Explosiveness", 28.7% variance) was characterised by sprint-related variables when accounting for the contribution of sprint frequency and sprint ratio. PC3 ("Agility / Accel-Decel Balance", 11.9% variance) was dominated by the acceleration-deceleration balance metric, representing agility-related movement characteristics.

TABLE 4.5: PCA Loadings for First Three Components

PCA Loadings

	Variable	PC1	PC2	PC3
hir_per_min_adj	hir_per_min_adj	-0.409	-0.248	0.143
sprint_per_min_adj	sprint_per_min_adj	-0.162	-0.520	-0.173
accels_per_min_adj	accels_per_min_adj	-0.377	0.044	-0.290
decels_per_min_adj	decels_per_min_adj	-0.410	0.121	0.129
distance_per_min_adj	distance_per_min_adj	-0.395	0.274	-0.220
intensity_ratio_adj	intensity_ratio_adj	-0.295	-0.382	0.217
sprint_ratio_adj	sprint_ratio_adj	-0.074	-0.558	-0.136
accel_decel_ratio_adj	accel_decel_ratio_adj	0.130	-0.070	-0.838
work_rate_adj	work_rate_adj	-0.403	0.215	-0.006
speed_endurance_adj	speed_endurance_adj	0.264	-0.262	0.180

Note: strong loadings (|loading| > 0.5). Component names: PC1 = Workload Volume, PC2 = Explosiveness, PC3 = Agility / Accel-Decel Balance.

4.4.4 PCA Biplot and Player Distribution

Figure 4.8 displays the distribution of players in the first two principal components and the contribution of performance variables to differences in position. The backs were clustered towards higher deceleration and work rate characteristics, the midfielders occupied a central region representing balanced outputs and the strikers aligned more strongly to sprint related variables. The variable orientation indicates that the sprint metrics had the greatest impact on PC2 and workload related measures dominated PC1.

FIGURE 4.8: PCA Biplot

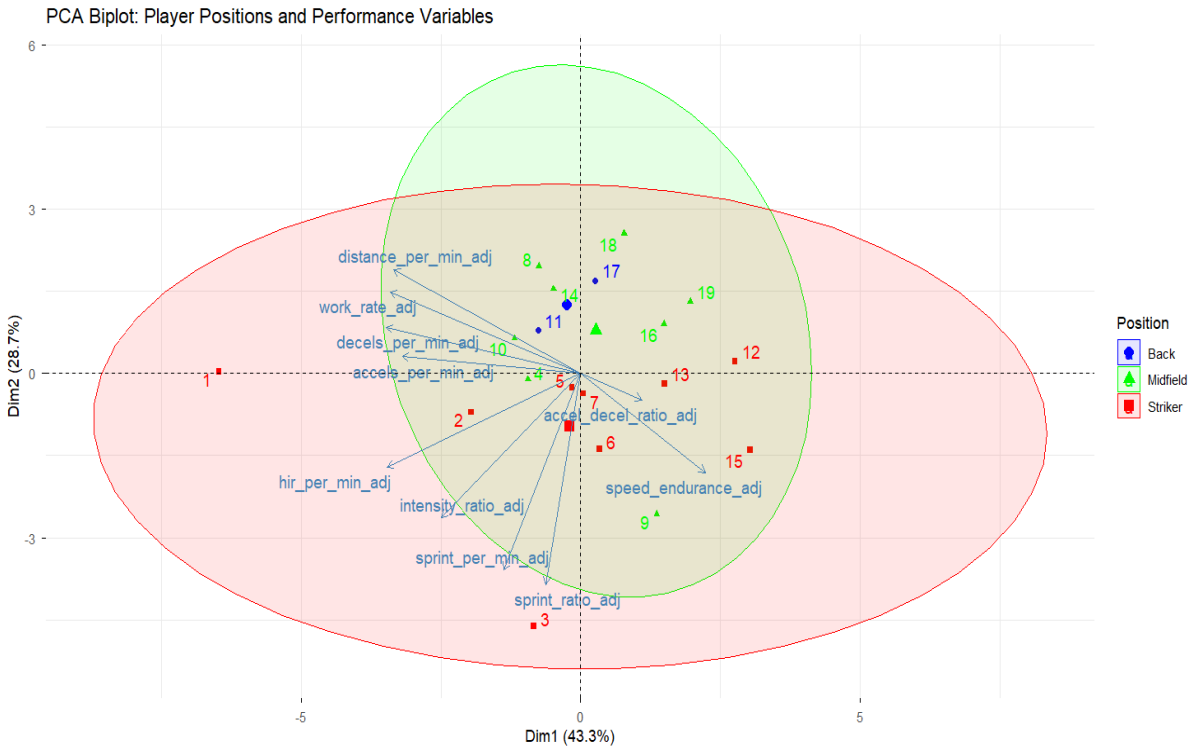


FIGURE 4.9: Player Distribution by Position (PC1 vs PC2)

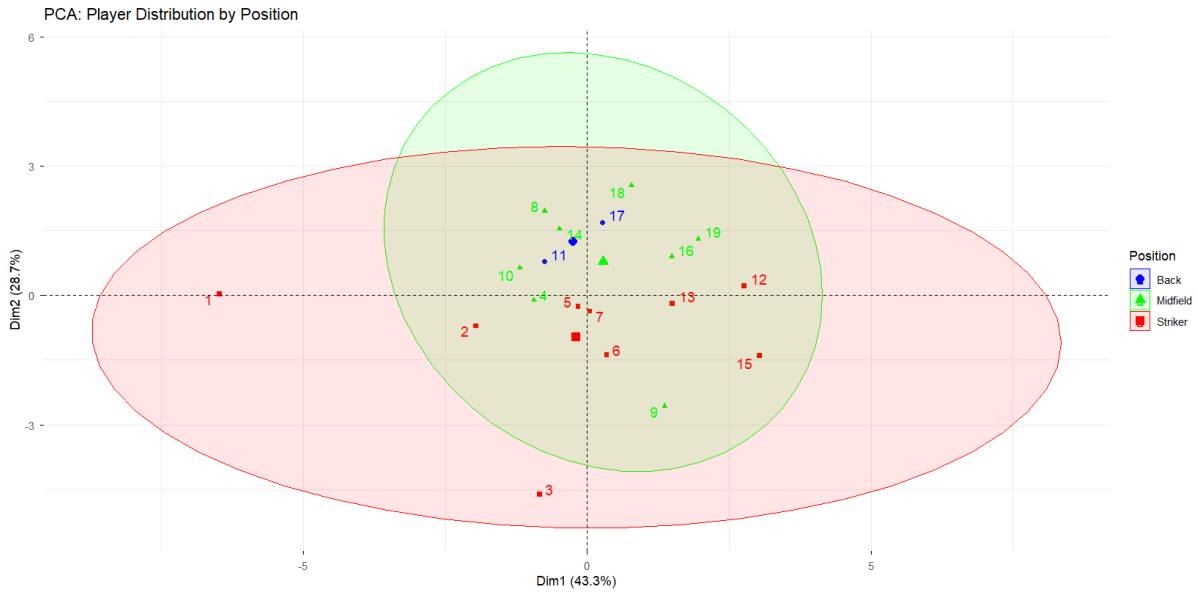


Figure 4.9 shows the distribution of players in the first two principal component space. Strikers tended to cluster in areas of a greater workload (low PC1) and explosiveness (low PC2). Midfielders were more spread, reflecting their mixed physical demands, backs more variable, partly due to the small sample size.

4.4.5 Component Scores Summary

Individual PCA scores of each participant are provided in Appendix C (Table C1). Further, the scores were applied in the process of clustering for identifying natural player archetypes.

4.5 CLUSTERING ANALYSIS

4.5.1 Determining Optimal Clusters

To define the natural group types among players, a k-means algorithm was applied to the scores for the first three components from the PCA. The most suitable cluster number was estimated using three different approaches: elbow approach (with minimum sum of squares), silhouette approach (with maximum average silhouette value), and the gap statistic method. As for the estimation results, all three techniques yielded exactly three clusters (k=3, Figure 4.10), further analysis was based on this result (Figure 4.10).

FIGURE 4.10: Optimal Cluster Determination

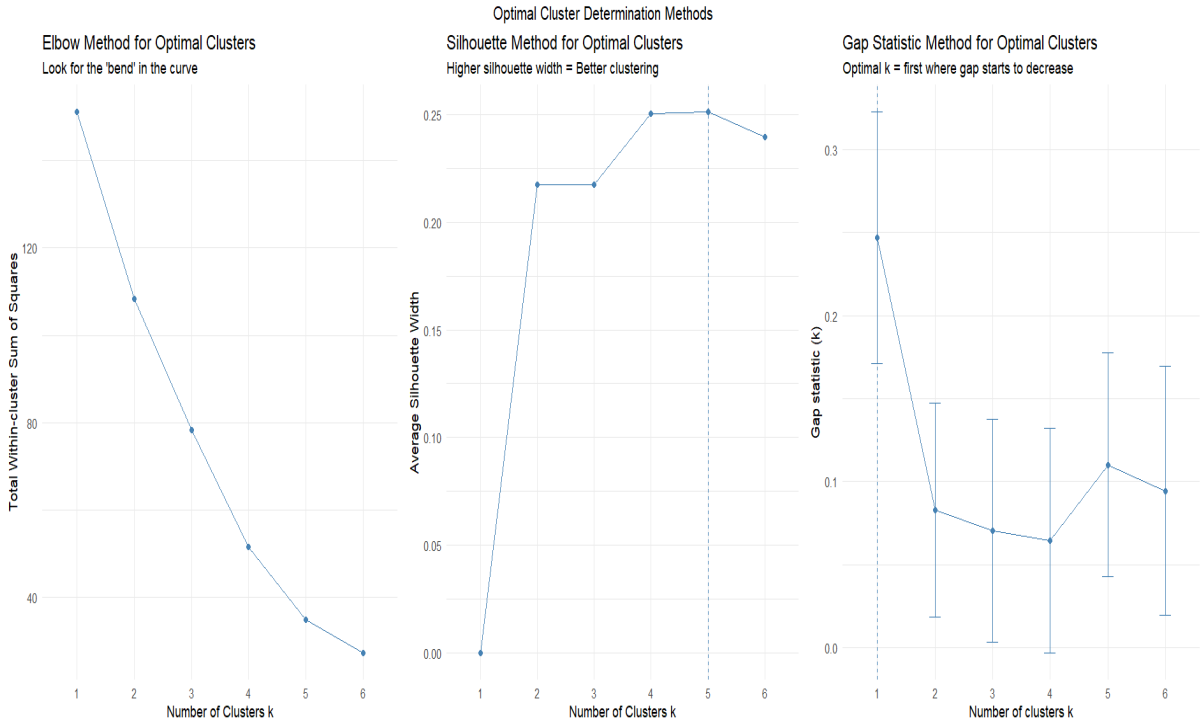


Figure 4.10 shows the techniques for identifying the best number of clusters. The Elbow criterion shows an elbow at k = 3, identifying this as the best solution. The Silhouette criterion also shows its maximum value at k = 3.

4.5.2 Cluster Sizes and Composition

Three clusters were identified, dividing the 19 players into groups of 6, 12 and 1. Cluster 1 consisted mainly of Strikers with one Midfielder, while Cluster 2 included two Backs, seven Midfielders and three Strikers. Cluster 3 comprised only one player (Angus Millar, Striker). It should be noted that Cluster 3 contains only one player (n = 1), which is methodologically unusual in k-means clustering and likely reflects the small sample size (n = 19). However, a look at his PCA scores and raw performance metrics showed consistently exceptional values, showing that this single-player cluster is a true high performer, rather than a technical artefact.

TABLE 4.6: Cluster Sizes and Positional Composition

Cluster Sizes and Positional Composition

Cluster	n	Backs	Midfielders	Strikers	Dominant_Position
1	6	0	1	5	Striker
2	12	2	7	3	Midfield
3	1	0	0	1	Striker

Table 4.7 presents the mean PCA component scores for each cluster, which informed the archetype interpretations.

TABLE 4.7: Cluster Profiles (Mean PCA Component Scores)

Cluster Profiles (Mean PCA Scores)

Cluster	Players (n)	Mean PC1	Mean PC2	Mean PC3	Positions
1	6	1.36	-1.66	0.29	Striker, Midfield
2	12	-0.14	0.83	-0.23	Striker, Midfield, Back
3	1	-6.47	0.03	1.07	Striker

Note: Lower PC1 indicates higher workload volume. Higher PC2 indicates greater explosiveness. Higher PC3 indicates better agility/accel-decel balance.

4.5.3 Player Archetypes

From the mean PCA component profiles, the three clusters were classified into distinctive player types (see Appendix D, Table D1). The first cluster consisted of 'Specialists' – players with high workloads but low explosiveness levels. In particular, the members of the first cluster scored higher in PC1 (1.36) but lower in PC2 (-1.66). This cluster included five players playing in the Striker position and one player in the Midfielder position. The second cluster consisted of 'Balanced Performers' . These players had scores close to average for all three components, meaning they have well-developed bodies and can play in various positions. Finally, Cluster 3 comprised an 'Elite Performer'. This player (Striker) scored very low on PC1 (-6.47), the lowest score among all players (indicating high workload volume), but relatively high on PC3 (1.07). Interpretation of these patterns Is discussed in Chapter 5.

FIGURE 4.11: Combined Cluster Visualization

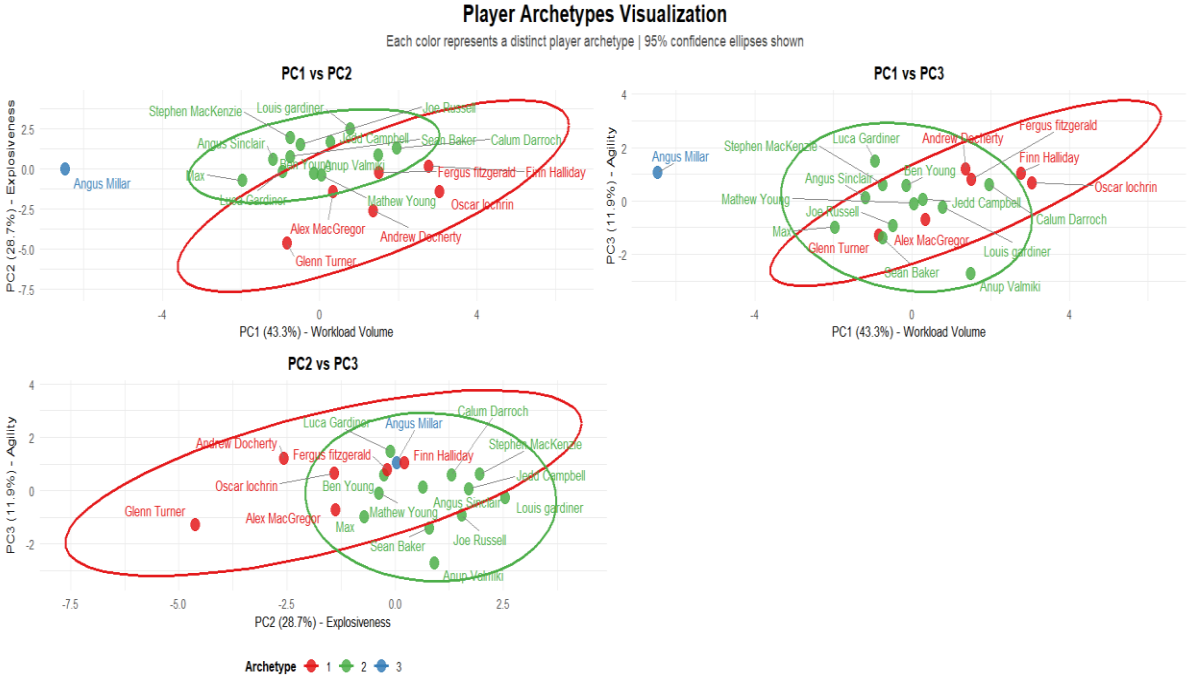


Figure 4.11 shows the three player types along the PCA dimensions. The elite performer, Angus Millar (Cluster 3), clearly stands out from others, especially with respect to PC1.

FIGURE 4.12: Radar Chart of Archetype Profiles

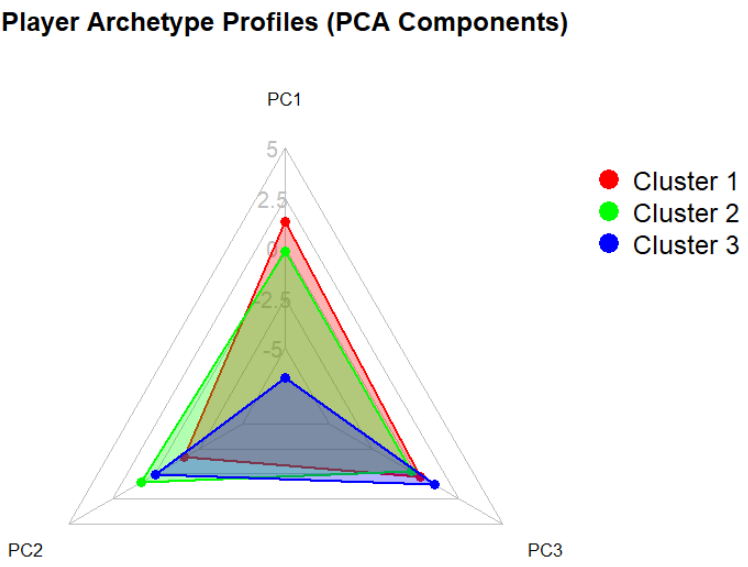


Figure 4.12 displays the average principal component scores of different player groups, representing the Specialist (high PC1 and low PC2), the Balanced Performer (close to mean in all components), and the Elite Performer (extremely low PC1 and high PC3).

4.5.4 Summary of Archetypes

These clusters produced three different types of player personas irrespective of their positions in the game. Cluster 1 consisted of five strikers and one midfielder. Cluster 2 had players of all the three positions with balanced physical attributes. Cluster 3 was a one-player cluster. This pattern is examined further in Chapter 5.

4.6 LINEAR DISCRIMINANT ANALYSIS (LDA)

4.6.1 LDA Classification Accuracy

LDA was conducted to analyze the performance of the three PCA components in predicting the playing positions. The accuracy of the classification model constructed using PC1, PC2, and PC3 was observed to be 73.7%.The pessimistic nature of LOOCV for small sample sizes (n = 19) is evident in the lowest accuracy obtained (57.9%) from cross-validation using leave-one-out. The true generalisation performance is likely to be between 57.9% and 73.7%. This point will be addressed further in Chapter 5. Given the small sample size, LOOCV tends to underestimate true model performance, so the cross-validated accuracy should be interpreted cautiously.

The per-class accuracy rate varied greatly depending on the playing position; it was highest for strikers (88.9%) and midfielders (75.0%). The Backs' accuracy was 0%; both Back subjects were misclassified as mid-fielders. It could have happened due to the following reasons: (1) too small number of samples belonging to this class (n=2 subjects); and (2) physical characteristics similarity of Backs and Midfielders based on their overlapping scores on PC2 (Explosiveness).

TABLE 4.8: LDA Classification Performance

Metric	Value
Overall Accuracy	73.68 %
Cross-Validation Accuracy	57.89%
Striker Accuracy	88.89% (8/9)
Midfield Accuracy	75.00% (6/8)
Back Accuracy	0.00 % (0/2)

4.6.2 Confusion Matrix

The confusion matrix (Table 4.9) shows that all eight Midfield players were classified as either Midfield or Striker. All nine Strikers were classified correctly or misclassified as Midfield. Both Back players were misclassified as Midfield, likely due to the limited sample size and physical similarities between Backs and Midfielders in this dataset.

TABLE 4.9: LDA Confusion Matrix

TABLE X: LDA Confusion Matrix

	Back	Midfield	Striker
Back	0	2	0
Midfield	0	6	2
Striker	0	1	8

FIGURE 4.13: LDA Confusion Matrix Heatmap

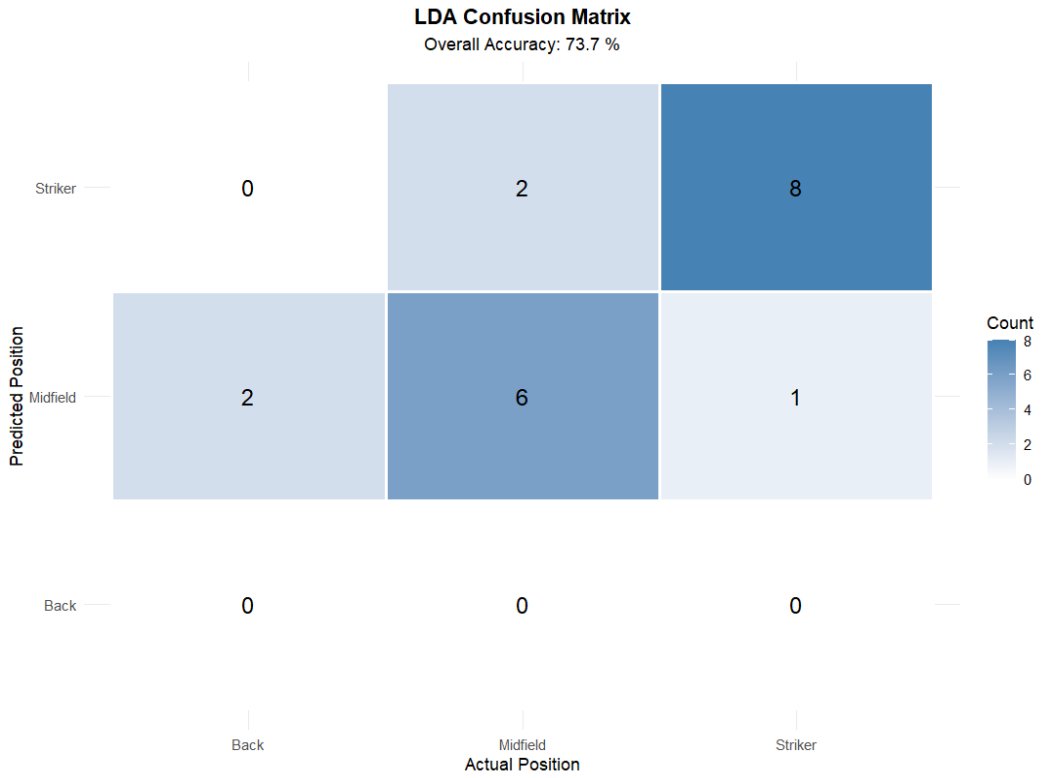


Figure 4.13: Heatmap for confusion matrix of LDA classification. The cells on the diagonal indicate accurate classifications.

4.6.3 Position Separation

As can be seen in Figure 4.14, LD1 was able to explain 93.3% of the total variance, whereas LD2 was able to explain 6.7% of the total variance. It is clear that Strikers (red) have been well-separated using LD1, which is due to the fact that PC2 (Explosiveness) had the highest loading on this LD1. There is much overlap between Midfielders (green) and Backs (blue), due to the physical similarity between these two positions.

FIGURE 4.14: LDA Position Separation Plot

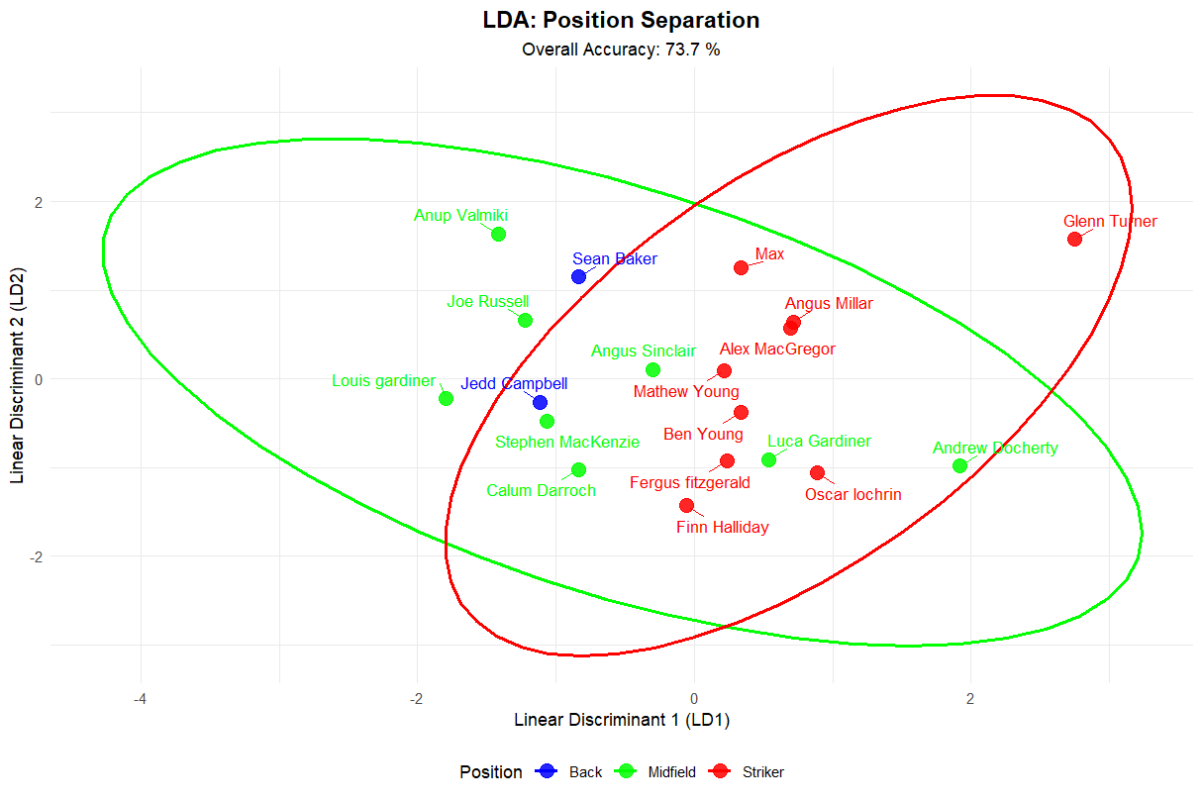


Figure 4.14 shows the LDA scatter plot, where LD1 (93.3% variance) distinguishes Strikers from the other groups, and LD2 (6.7% variance) further aids the separation.

4.6.4 Variable Importance

Table 4.10 shows the discriminant function coefficients. LD1 accounted for 93.3% of the between-group variation, with PC2 as the most influential component. LD2 accounted for 6.7% of the variation, with PC3 contributing most strongly. It can thus be concluded that sprint capacity differentiates between playing positions in field hockey.

TABLE 4.10: Discriminant Function Coefficients

Discriminant Function Coefficients

Variable	LD1	LD2
PC1	-0.0703	-0.2244
PC2	-0.6574	-0.0906
PC3	0.2664	-0.7562

Note: Larger absolute coefficients indicate stronger contribution to the discriminant function. PC2 contributes most to LD1 (separating Strikers from others). PC3 contributes most to LD2 (separating Backs and Midfielders).

4.6.5 Misclassified Players

Five instances of incorrect classification were seen in the results of applying the LDA model (see full table in Appendix D, Table D2). There were two cases of Midfielders (Luca Gardiner, Andrew Docherty) being classified as Strikers. Interestingly, both of these players had higher PC3 values (> 1.2) for Agility / Accel-Decel Balance, explanation for these trends are discussed in Chapter 5. Two Backs (Sean Baker, Jedd Campbell) were incorrectly classified as Midfielders because of physical similarities between Backs and Midfielders and One Striker (Finn Halliday) was misclassified as Midfielder, because his PC2 (Explosiveness) score (-0.71) was lower compared to other Strikers. Misclassification patterns should be interpreted in the context of the small and imbalanced sample across positions.

4.6.6 Comparison with Cluster-Based LDA

To make a comparison, an LDA model was also developed using the clusters (obtained using PCA scores) as the only predictor variable for playing positions (Table 4.11). The model obtained a prediction rate of 68.4%, which is slightly inferior to the prediction rate of 73.7% obtained from the PCA component model. The difference in prediction rate implies that the PCA components contain more discriminating power than the cluster labels.

TABLE 4.11: Model Comparison – PCA Components vs Clusters

Model Comparison – PCA Components vs Clusters

Method	Accuracy
PCA Components (PC1, PC2, PC3)	73.68%
Cluster Assignments	68.42%

4.7 DECISION TREE ANALYSIS

4.7.1 Decision Tree Performance

A decision tree was constructed with the three PCA components as predictors of playing position, with an aim to test their ability to predict position based on the physical dimensions they represent. The tree performed very well in predicting positions within the training data set, reaching an accuracy level of 94.7%, with only one player out of 19 misclassified (Table 4.12). In contrast, however, the accuracy of the tree when tested via cross-validation was much poorer at 68.7%. This significant difference in results may be explained by the possible occurrence of overfitting in this small sample (n=19). Decision trees are sensitive to small datasets, so the training accuracy and split structure should be interpreted with caution.

TABLE 4.12: Decision Tree Performance

Metric	**Value**
Training Accuracy	94.74%
5-fold Cross-Validation Accuracy	68.67%
LDA Accuracy (for comparison)	73.68%

Note: The huge difference between training and validation accuracy values signifies overfitting. Validation accuracy value (68.7%) can be used as a more accurate measure for assessing the model's generalization capability.

4.7.2 Confusion Matrix and Misclassifications

Based on the confusion matrix (Table 4.13), it can be seen that the decision tree model was successful in correctly classifying all Strikers (n=9) and Backs (n=2). One of the Midfielders (Joe Russell) was erroneously classified as a Back. The relatively high sensitivity scores among Strikers and Backs reflect differences in their physical characteristics, particularly their PC2 (Explosiveness) values. One of the Midfielders with PC2 score of 1.54 and PC3 score of -0.91 was incorrectly classified as a Back.

TABLE 4.13: Decision Tree Confusion Matrix

Decision Tree Confusion Matrix

Actual	Back	Midfield	Striker
Back	2	0	0
Midfield	1	7	0
Striker	0	0	9

FIGURE 4.15: Decision Tree Confusion Matrix Heatmap

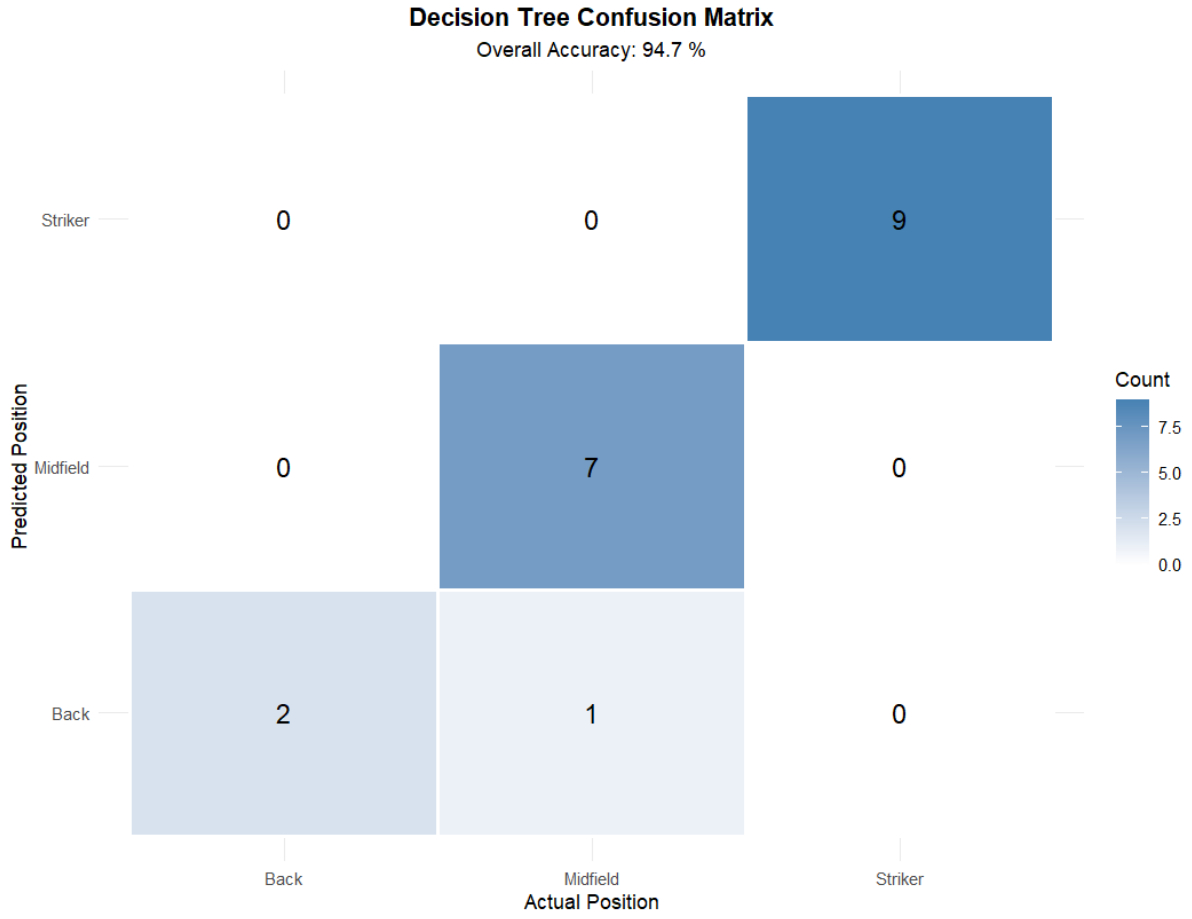


Figure 4.15 shows the decision tree confusion matrix heatmap. Diagonal elements indicate correctly classified instances, with only one incorrect prediction (a midfielder classified as a back).

FIGURE 4.16: Decision Tree Variable Importance

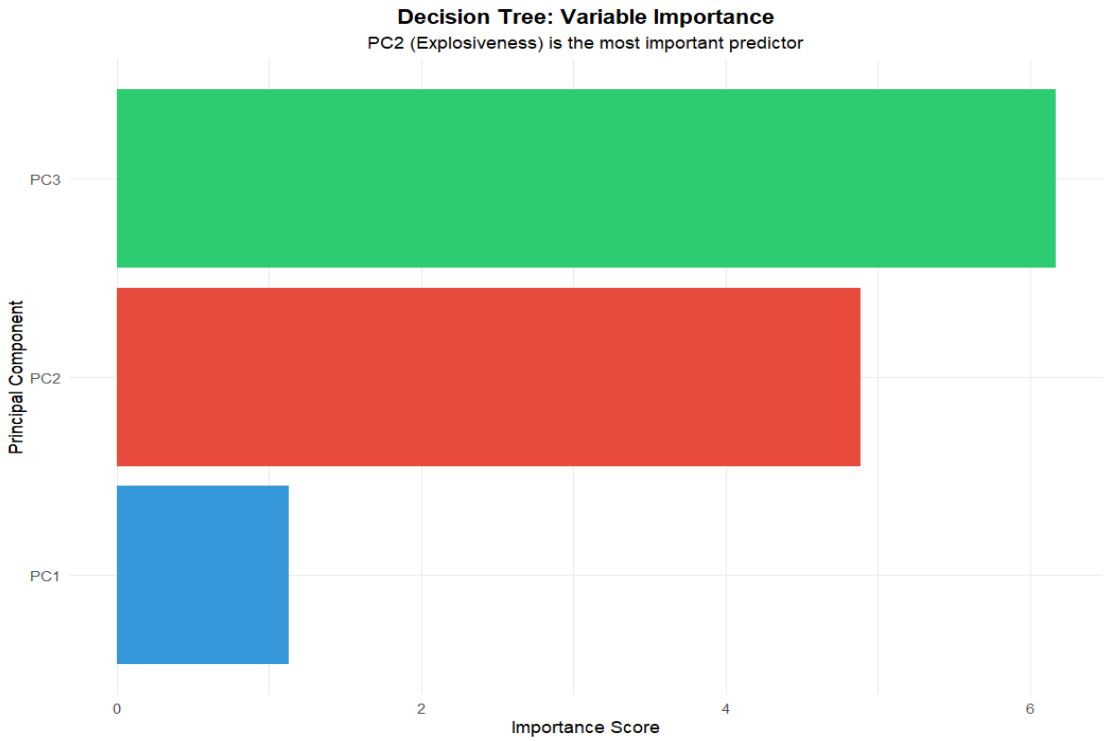


Figure 4.16: Decision tree variable importance chart. PC3 (agility) is the strongest predictor (importance 6.17) followed by PC2 (explosiveness; importance 4.89). Contribution of PC1 (workload volume) into prediction is almost negligible.

4.7.3 Decision Tree Structure and Rules

From the tree structure diagram shown in Figure 4.17 above, one is able to understand the decision rules behind the decision tree model for position classification. In this case, the decision tree model first splits by PC2 with a cut-off value of 0.425. In the case where the PC2 is ≥ 0.425, the players move to the right branch, whereby splits on PC1 and PC3 determine whether such players are Midfielders or Backs. In cases where the value of PC2 is less than 0.425, then the players move to the left branches, whereby a split on PC3 with a threshold value of 1.137 distinguishes between Midfielders and Strikers.

The decision tree rules include the following:

Decision Rule 1	Where PC2 < 0.425 and PC3 < 1.137 → Striker (n=9)
Decision Rule 2	Where PC2 < 0.425 and PC3 ≥ 1.137 → Midfielder (n=2)
Decision Rule 3	Where PC2 ≥ 0.425, PC1 < 0.522, and PC3 < 0.103 → Back (n=3)
Decision Rule 4	Where PC2 ≥ 0.425, PC1 < 0.522, and PC3 ≥ 0.103 → Midfielder (n=2)
Decision Rule 5	Where PC2 ≥ 0.425 and PC1 ≥ 0.522 → Midfielder (n=3)

FIGURE 4.17: Decision Tree Diagram

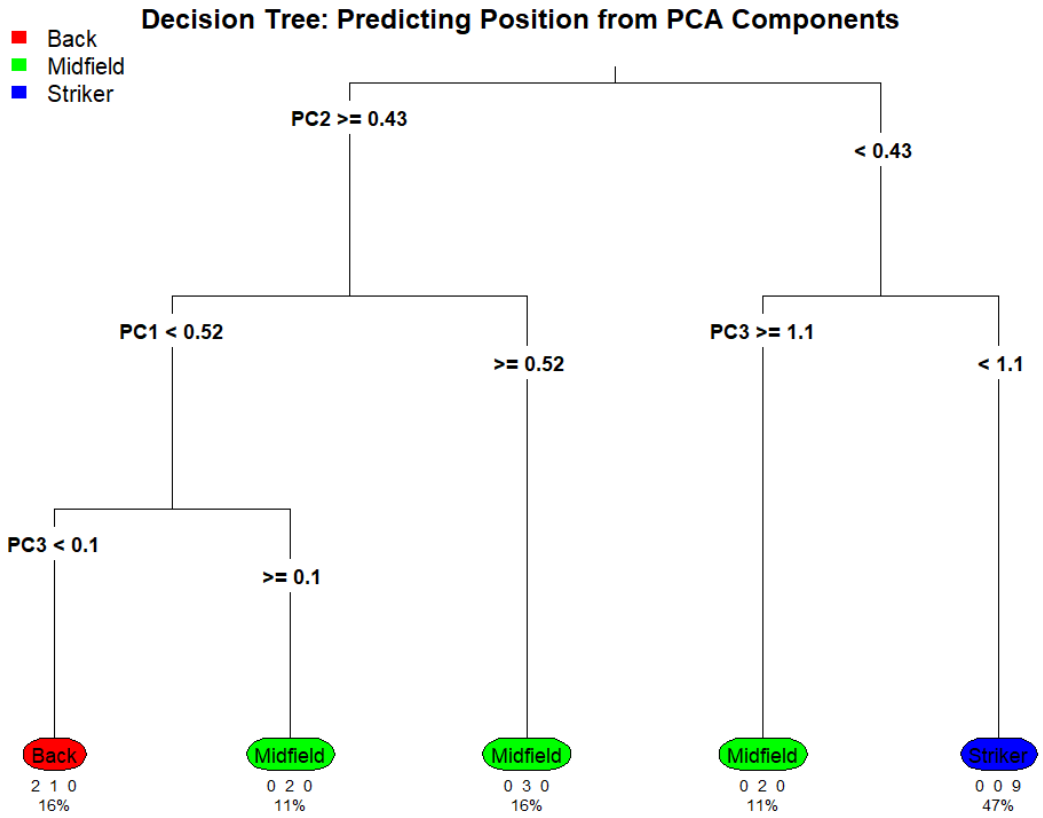


Figure 4.17 shows the hierarchical splitting rules in the decision tree. PC2 is used for the first split, followed by splits on PC1 and PC3.

FIGURE 4.18: Decision Boundaries (PC1 vs PC2)

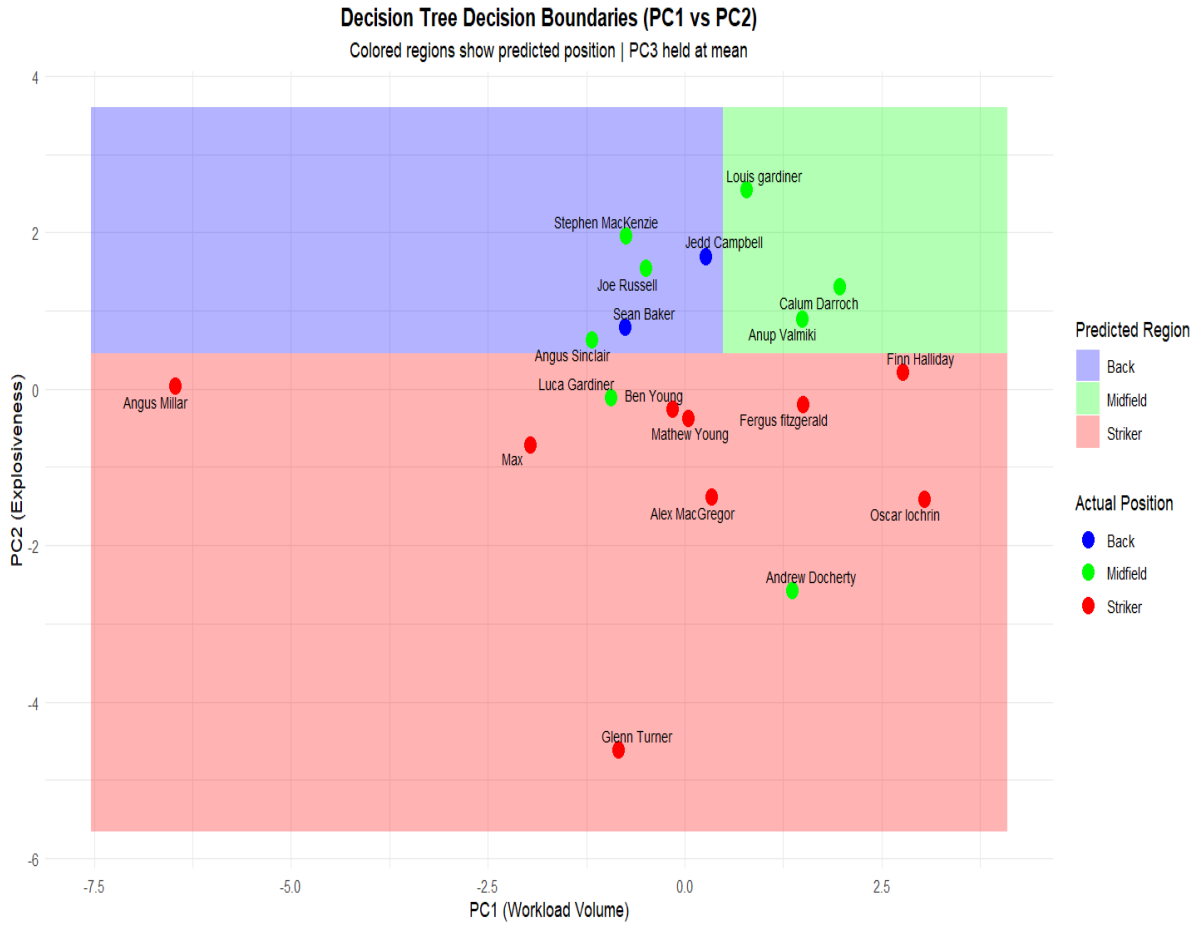


Figure 4.18: Classification boundaries of the decision tree in the space defined by PC1 and PC2 (PC3 fixed at the average). Coloured regions correspond to different positions predicted to occupy the corresponding areas. Points are actual players plotted according to their positions in the space and coloured according to their names. A boundary drawn at PC2 ≈ 0.425 differentiates Strikers (to the left) and Midfielders/Backs (to the right).

4.7.4 Variable Importance and Model Comparison

As revealed in Table 4.14, variable importance measures revealed that the predictor that carried the most weight was PC3, known as Agility (importance = 6.17), followed closely by PC2, which represents Explosiveness (importance = 4.89). Variable PC1 represented workload volume, which had no significance in discriminating between playing positions (importance = 1.13).

TABLE 4.14: Decision Tree Variable Importance and Model Comparison

Variable	Importance
PC3 (Agility	6.17
PC2 (Explosiveness)	4.89
PC1 (Workload Volume)	1.13

A comparison between all the classification models is presented in Table 4.15 and Fig. 4.19. While the decision tree was more accurate during training (94.7%) compared to LDA based on PCA component reduction (73.7%) and LDA based on clusters (68.4%), All models performed substantially better than the random baseline (33.3%). However, the big discrepancy between the performance in training (94.7%) and cross-validation (68.7%) accuracy indicates the presence of overfitting in case of the decision tree. Considering that there is small sample size's observations, LDA (linear models) could yield more accurate estimates of classification accuracy in the whole population.

TABLE 4.15: Model Performance Comparison

Model	Type	Accuracy
Decision Tree	Non-Linear	94.74%
LDA (PCA Components)	Linear	73.68%
LDA (Clusters)	Linear	68.42%
Random Baseline (Chance)	Baseline	33.33%

Note: Decision Tree accuracy reflects training performance; cross-validation accuracy (68.7%) is a more conservative estimate.

FIGURE 4.19: Model Performance Comparison

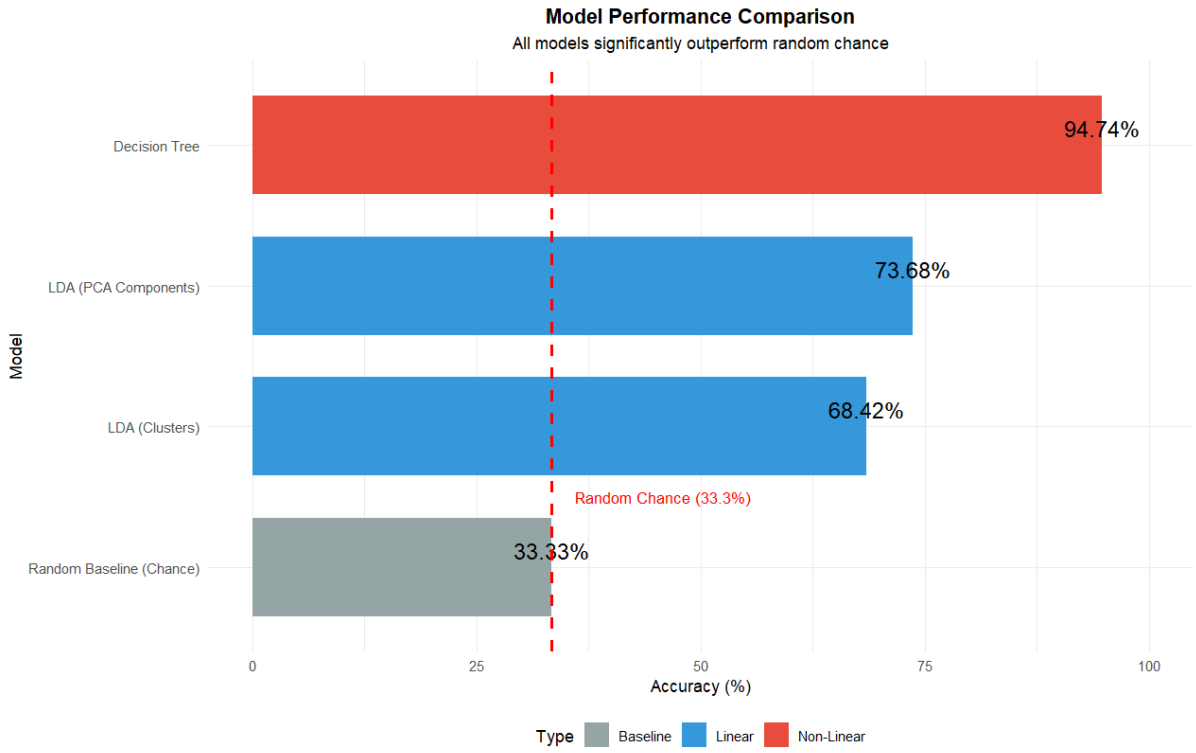


Figure 4.19 shows that the decision tree was the best-performing model with an accuracy of 94.7%, followed by LDA with PCA components (73.7%) and LDA using clusters (68.4%). All models significantly outperformed random assignment (33.3%).

CHAPTER 5: DISCUSSION

5.1 Summary of Key Findings

To investigate if data-based physical archetypes in elite men's hockey are consistent with or different from traditional positional roles, this study used a multi-method analytical pipeline that applied mixed effects models, PCA, clustering, and classification [8], [9]. This discussion is based on three findings. initial, PC2 (Explosiveness), which performed the first split of the decision tree and was responsible for 93.3% of the variation across groups in LDA, was found to be the main discriminator of playing position [11]. Second, three archetypes were identified: an Elite Performer (n = 1), Balanced Performers (n = 12), and Specialists (n = 6). Third, participants from different positions formed clusters, and players from the same position separated among clusters, showing high within-position diversity [6]. Collectively, these results indicate that the physical characteristics of elite hockey players are not amenable to positional classification [1], [3].

5.2 PC2 (Explosiveness) as the Primary Position Discriminator

5.2.1 Why PC2 Is the Most Substantive Result

PC2 (Explosiveness) was the most consistent predictor of playing position in all analysis techniques [11]. In the LDA model, PC2 explained 93.3% of the variance among groups, and had the highest coefficient on LD1 (–0.657). The first split in the decision tree was at a threshold of 0.425 on PC2, which suggests that the most important decision rule for predicting position is whether or not a player's explosiveness is above this value.

This finding suggests a major component of positional play in elite men's hockey: The key physical differentiator of strikers from other positions is the ability to sustain repeated high-intensity sprints [2]. In contrast to this, midfielders and backs have similar physical profiles, with only differences in secondary dimensions such as agility (PC3) or workload volume (PC1). This interpretation is backed up by the LDA confusion matrix with a poorer accuracy in midfielders (75.0%) and backs (0%) than in strikers (88.9%) due to a smaller physical distinction between these two roles.

5.2.2 Why Explosiveness Differentiates Positions

Positions are inherently explosive because elite hockey is so tactical. For example, strikers often finish scoring opportunities, attack space and create separation using short, maximum sprints [1]. While backs rely more on positional organisation, recovery runs and change of direction actions, midfielders run at a higher intensity, continuously over longer distances [3]. The physical data is a mirror of these functional differences: Midfielders covered the most distance overall, backs made the most number of accelerations and decelerations, whereas strikers had the highest HIR and sprint outputs [1].

5.2.3 Comparison with Previous Research

This finding is in line with previous research that has reported that forwards in invasion sports (e.g., hockey) consistently produce the greatest sprint outputs [1], [2]. The present study extends previous work and demonstrates explosiveness to be position-specific and the most important predictor of positional categorisation in both linear and nonlinear models [11]. Nevertheless, the agreement between LDA (73.7% accurate) and decision tree cross validation (68.7% accurate) gives more confidence in this conclusion despite the small sample size [36].

5.3 Alignment and Divergence Between Archetypes and Positions

5.3.1 Alignment: Specialists and Strikers

The Specialist archetype (Cluster 1) was mostly composed of Strikers (5/6 players), showing a degree of partial congruence between positions and data-driven archetypes. These players were less explosive than the team average (negative PC2 scores) and had less workload volume (positive PC1 values). This suggests that these strikers may rely more on tactical positioning, movement patterns or finishing ability rather than repeatedly running [6]. The alignment of Specialists and Strikers makes it obvious that there are some physical characteristics that go with some roles.

5.3.2 Divergence: Balanced Performers Across All Positions

The largest and most heterogeneous archetype was the Balanced Performer (Cluster 2) with players from all three positions (two defenders, seven midfielders and three

attackers). This result shows that most of the players in this team possess a similar physical profile, which does not significantly differentiate the roles [4]. For these players, the technical ability, the tactical awareness and/or the coach preference probably have more to do with where they play than just the physical attributes.

The fact that there are three strikers in the balanced group suggests not all strikers are explosive, some can use other tactical or physical techniques. Likewise, the fact that backs and midfielders are grouped together suggests that these positions are not physically distinct on this team [1].

5.3.3 What This Reveals About Positional Function

These trends suggest that elite men's hockey positions are more tactical responsibilities than strict physical classifications. Explosiveness is what separates strikers as a category, but there is a lot of individual variance within any position. A midfielder can be more agile or have a large work rate, a striker can be explosive or good at possession, and a back can have the same physical characteristics as a midfielder. This subtlety is captured by the multi-method pipeline, revealing organic groups beyond the conventional positional labelling [7].

5.4 The Elite Performer Cluster: Insightful but Not Generalisable

The Elite Performer cluster (Cluster 3) consisted of a single player (Angus Millar), who demonstrated exceptionally high workload volume (PC1 = –6.47) and above-average agility (PC3 = 1.07). While analytically interesting, this cluster must be interpreted with caution. A single-player cluster may reflect:

- a genuine elite profile,
- a sample-specific outlier,
- measurement variation, or
- an artefact of the small sample size (n=19), [17].

No generalisations can be made about the "Elite Performer" type. This is a finding that is hypothesis-generating not confirmatory. The proper interpretation is that this athlete represents an outlier in this team.

5.5 Methodological Considerations

5.5.1 KMO Issue

The KMO value (0.443) was below the recommended value of 0.6 [19], indicating that the correlation structure was not suitable for PCA. Bartlett's test was significant, although some pairs of variables showed high correlations and sampling adequacy was reduced, which supports the use of PCA as a reduction of dimensions technique [8].

5.5.2 Decision Tree Overfitting

The decision tree was significantly overfitted (training accuracy 94.7% versus 68.7% cross validation). This is expected with small sample sizes [11]. The LDA results were more robust and should therefore be weighted more for interpretation.

5.6 Methodological Strengths and Limitations

5.6.1 Strengths

There are several methodological features of this study that increase the interpretability and robustness of the findings. The combination of mixed effects modelling, PCA, clustering and classification into one analytical pipeline is a new and comprehensive method for profiling players in elite hockey [9]. Firstly, Each approach is tuned to a different analytical challenge: clustering uncovers the latent archetypes; PCA alleviates multicollinearity; mixed models allow for contextual heterogeneity; LDA/decision trees serve to assess the predictive ability of these archetypes. The successive use of these techniques offers a coherent framework that is rarely used in the field of hockey research.

Second, a methodological innovation is the use of mixed effects models to pre-process the GPS data prior to the PCA. Removing variance associated with session type and season month, the PCA components more faithfully represent fundamental player traits rather than artefacts of context. This modification is important when the data is obtained from single team datasets with unbalanced exposure to games and training sessions [18].

Third, the study conducted thorough assumption testing during the analytical process. We used ICC estimates [16], KMO and Bartlett's tests for PCA suitability, residual normality checks for mixed models and cross validation for classification models as diagnostics [36]. Such openness allows for the critical evaluation of the methodological appropriateness of each analytical step.

Finally, the dataset consists of 397 session-level observations recorded over a competitive season, thereby offering ecologically valid insights into patterns of elite performance [5]. The addition of rate-based feature engineering adds to the analysis by making features comparable across sessions of different lengths [31].

5.6.2 Limitations

Despite these strengths, several limitations must be acknowledged, each with implications for the interpretation and generalisability of the findings.

Sample size and cluster stability

The main limitation is the small sample size (n = 19 players). This leaves room for sample variability to influence PCA loadings and classification accuracy. The Elite Performer cluster contains only one player, which limits its generalisability and prevents it from being treated as a verified archetype [17]. However, this player's consistently exceptional performance across all analyses suggests that the single-player cluster reflects a genuine high-performer rather than a statistical artefact.

Positional imbalance

The dataset only contains 2 Backs, so it is difficult to make any reliable inference for this position. This extreme imbalance in position means that the 0% LDA classification accuracy for Backs is to be expected, but is not analytically meaningful, as LDA cannot form a stable discriminant boundary with only two cases. Hence, the misclassification of Backs is a limitation in the data rather than a weakness in the modelling, decreasing the representativeness of the archetypal position correlations [1].

PCA sampling adequacy

The modest KMO value (0.443) indicates limited sampling adequacy for PCA, meaning that the component structure may be less stable than ideal. Despite Bartlett's test being significant and there being several pairs of variables with high correlations [8], the low KMO suggests that the PCA results should be interpreted with caution. Future studies with larger samples are needed to confirm whether the same component structure emerges.

Mixed model residual normality

Residuals of the mixed model were not normally distributed (Shapiro–Wilk $p < 0.001$). This is still a drawback even if mixed models are robust to non-normality with sample sizes larger than 30 sessions (not players) [10]. The non-normality is the skewed distribution typical of elite performance data, where extreme values tend to be prevalent, and this violation may affect the accuracy of fixed-effect estimations even though mixed models are generally robust to such issues.

Decision tree overfitting

The decision tree exhibited significant overfitting, with a large gap between training accuracy (94.7%) and cross validation accuracy (68.7%). This makes the tree-based decision rules less reliable, but it is expected for small data sets [11]. Thus the LDA results should be given higher weight as they were more stable.

Single-team, single-season design

The findings reflect the characteristics of one elite men's team in one season of competition. Thus, there is limited external validity and the archetypes found may not generalise to other teams, levels of competition or female athletes.

5.7 Recommendations for Future Research

Future research should build on the methodological foundations of this study while addressing its limitations.

Multi-team, larger-scale replication

The most important next step is to replicate this analytical procedure in several teams and larger samples (n ≥ 50–100). PCA sampling would be more appropriate, cluster solutions would be more stable, overfitting in classification models would be reduced, archetypes could be identified more accurately, and with a larger dataset. And on top of that it would give us the opportunity to explore whether the Elite Performer profile is a fluke of this sample or a real archetype [7].

Improved representation of defensive positions

Future research should focus on recruiting more back players to allow for more accurate models of positional difference. This would clarify whether the physical resemblance between midfielders and backs seen in this study is a consequence of positional imbalance or a real characteristic of top hockey [1].

Alternative dimensionality reduction techniques

Given the low KMO value indicates that in the future, researcher should consider other techniques like factor analysis t SNE or UMAP [37] and compare them with PCA. These methods might be able to identify nonlinear co-relationships between performance factors that PCA cannot identify.

Use of ensemble classification methods

Upcoming research may explore ensemble methods like Random Forests, Gradient Boosting or XGBoost to prevent overfitting. Besides, these models help determine the true predictive power of physical components and typically they perform better than single decision trees for generalization[38].

Longitudinal and cross-context validation

The pipeline could be scaled to different seasons, age groups, and competitive levels, and archetype stability could be studied over time and across contexts. The results would have an even wider range of applicability if the study were expanded to include female and young athletes.

Integration of technical and tactical data

Finally, future research should include technical skill metrics, tactical tracking data, and contextual factors (e.g., opposition strength, match phase). Such an approach would allow for a more complete understanding of how technical, tactical and physical aspects co-shape positional duties and player archetypes [6].

CHAPTER 6: CONCLUSION

6.1 Summary of Key Findings

This study employed a multi-method analytical pipeline of Linear Mixed Models, Principal Component Analysis, K-means clustering, Linear Discriminant Analysis, and Decision Trees to profile elite men's hockey players.

The most important finding was that PC2 (Explosiveness/Sprint Capacity), explaining 93.3% of the between group variance of LDA and the first split of the decision tree at a cut-off of 0.425 was the primary discriminator of playing position.

We identified three player archetypes: Elite Performer (n = 1, very high workload volume), Balanced Performers (n = 12, players from all three positions, average workload volume and higher explosiveness) and Specialists (n = 6, predominantly Strikers, lower workload volume and lower explosiveness). These archetypes showed a large within-position variation, both in terms of conforming to and deviating from traditional positions.

The PCA reduced ten factors to three components accounting for 84.0% of the total variance: Agility (11.9%), Explosiveness (28.7%), and Workload Volume (43.3%).

6.2 Answers to Research Questions

PCA identified three underlying physical dimensions: explosiveness (PC2, 28.7%), workload volume (PC1, 43.3% of variance) and agility (PC3, 11.9%). ICC of 0.270 indicated moderate player-level clustering and between-player differences accounted for 27% of variance. The most positive deviation (+3.30) was for Angus Millar.

LDA predicted playing position with 73.7% accuracy due to the small sample size with strikers being the best categorised (88.9%) and backs being completely misclassified (0%). Decision tree training accuracy (94.7%) dropped to 68.7% with cross-validation, indicating overfitting.

6.3 Contribution to Knowledge

This study represents that a multi-method analytical pipeline can be leveraged to identify important, data-driven archetypes in top men's hockey. It implies that conventional positional categories fail to encompass the diversity of physiological profiles within a team and that the ability to sprint emerges as the main physical characteristic differentiating player positions, rather than the overall workload. This research offers a method that can be repeated to relate the data on physical performance to the positional roles and archetype systems by integrating mixed models PCA clustering and classification into one system.

6.4 Practical Implications

The findings imply a number of practical implications for coaches, performance analysts and support staff. Firstly, even though midfielders and backs may have similar physical profiles , the planning of position-specific training must emphasize the sprinting ability and its importance In particular for strikers. Secondly, training and monitoring should not be based on averages of position but should be individually tailored by identifying the specialists, the Balanced Performers and an Elite Performer. Finally, the misclassification of Backs and the presence of an Elite Performer outlier exemplify the need for continued assessment of workload management, pathways for talent development, and role allocation within elite teams.

6.5 Final Statement

This study reveals that positions in men's elite hockey at top level are essentially tactical functions which may be achieved with different physical methods rather than being defined by fixed physical categories. The largest physical difference between positions is sprint capacity (PC2), which indicates the need for explosiveness in position specific training. The three data-driven archetypes provide a warning against over-reliance on positional averages in applied settings as they demonstrate large within-position variance and overlapping physical profiles.

Although the Elite Performer category is quite significant statistically, it cannot be reached from one player alone. There is a need for future studies to verify the archetypal formation, enhance generalisability and device data based techniques to player profiling in top level hockey, through the use of larger, multi- team datasets.

REFERENCES

[1] D. Jennings, S. J. Cormack, A. J. Coutts, and R. J. Aughey, “GPS Analysis of an International Field Hockey Tournament,” *Int. J. Sports Physiol. Perform.*, vol. 7, no. 3, pp. 224–231, Sep. 2012, doi: 10.1123/ijspp.7.3.224.

[2] J. D. Vescovi, “Locomotor, Heart-Rate, and Metabolic Power Characteristics of Youth Women’s Field Hockey: Female Athletes in Motion (FAiM) Study,” *Res. Q. Exerc. Sport*, vol. 87, no. 1, pp. 68–77, Jan. 2016, doi: 10.1080/02701367.2015.1124972.

[3] Morencos, E., “Kinematic demands of international competition in women’s field hockey,” 2019, [Online]. Available: https://www.redalyc.org/journal/5516/551660713006/551660713006.pdf

[4] P. S. Bradley and T. D. Noakes, “Match running performance fluctuations in elite soccer: Indicative of fatigue, pacing or situational influences?,” *J. Sports Sci.*, vol. 31, no. 15, pp. 1627–1638, Nov. 2013, doi: 10.1080/02640414.2013.796062.

[5] J. J. Malone, R. Lovell, M. C. Varley, and A. J. Coutts, “Unpacking the Black Box: Applications and Considerations for Using GPS Devices in Sport,” *Int. J. Sports Physiol. Perform.*, vol. 12, no. s2, pp. S2-18-S2-26, Apr. 2017, doi: 10.1123/ijspp.2016-0236.

[6] R. Rein and D. Memmert, “Big data and tactical analysis in elite soccer: future challenges and opportunities for sports science,” *SpringerPlus*, vol. 5, no. 1, p. 1410, Dec. 2016, doi: 10.1186/s40064-016-3108-2.

[7] C. T. Woods, J. Veale, J. Fransen, S. Robertson, and N. F. Collier, “Classification of playing position in elite junior Australian football using technical skill indicators,” *J. Sports Sci.*, vol. 36, no. 1, pp. 97–103, Jan. 2018, doi: 10.1080/02640414.2017.1282621.

[8] Jolliffe, I. T., *Principal Component Analysis*. in Springer Series in Statistics. New York: Springer-Verlag, 2002. doi: 10.1007/b98835.

[9] D. Bates, M. Mächler, B. Bolker, and S. Walker, “Fitting Linear Mixed-Effects Models Using **lme4**,” *J. Stat. Softw.*, vol. 67, no. 1, 2015, doi: 10.18637/jss.v067.i01.

[10] H. Schielzeth *et al.*, “Robustness of linear mixed-effects models to violations of distributional assumptions,” *Methods Ecol. Evol.*, vol. 11, no. 9, pp. 1141–1152, Sep. 2020, doi: 10.1111/2041-210X.13434.

[11] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*. in Springer Series in Statistics. New York, NY: Springer New York, 2009. doi: 10.1007/978-0-387-84858-7.

[12] R. J. Aughey, “Applications of GPS Technologies to Field Sports,” *Int. J. Sports Physiol. Perform.*, vol. 6, no. 3, pp. 295–310, Sep. 2011, doi: 10.1123/ijspp.6.3.295.

[13] J. Z. Lim, A. Sim, and P. W. Kong, “Wearable Technologies in Field Hockey Competitions: A Scoping Review,” *Sensors*, vol. 21, no. 15, p. 5242, Aug. 2021, doi: 10.3390/s21155242.

[14] L. S. Luteberget and M. Gilgien, “Validation methods for global and local positioning-based athlete monitoring systems in team sports: a scoping review,” *BMJ Open Sport Exerc. Med.*, vol. 6, no. 1, p. e000794, Aug. 2020, doi: 10.1136/bmjsem-2020-000794.

[15] B. Romero-Moraleda, E. Morencos-Martínez, L. Torres-Ronda, and D. Casamichana, “Analysis of congested schedule on competition external load in field hockey. [Análisis de la carga externa de competición en un periodo de congestión en hockey hierba].,” *RICYDE Rev. Int. Cienc. Deporte*, vol. 16, no. 60, pp. 143–152, Apr. 2020, doi: 10.5232/ricyde2020.06003.

[16] T. K. Koo and M. Y. Li, “A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research,” *J. Chiropr. Med.*, vol. 15, no. 2, pp. 155–163, Jun. 2016, doi: 10.1016/j.jcm.2016.02.012.

[17] W. G. Hopkins, “Measures of Reliability in Sports Medicine and Science:,” *Sports Med.*, vol. 30, no. 1, pp. 1–15, 2000, doi: 10.2165/00007256-200030010-00001.

[18] S. Nakagawa and H. Schielzeth, “A general and simple method for obtaining R^2 from generalized linear mixed-effects models,” *Methods Ecol. Evol.*, vol. 4, no. 2, pp. 133–142, Feb. 2013, doi: 10.1111/j.2041-210x.2012.00261.x.

[19] H. F. Kaiser, “An Index of Factorial Simplicity,” *Psychometrika*, vol. 39, no. 1, pp. 31–36, Mar. 1974, doi: 10.1007/BF02291575.

[20] D. Ulloa-Díaz *et al.*, “Exploring body composition and physical condition profiles in relation to playing time in professional soccer: a principal components analysis and Gradient Boosting approach,” *Front. Physiol.*, vol. 16, p. 1659313, Oct. 2025, doi: 10.3389/fphys.2025.1659313.

[21] C. De La Fuente *et al.*, “Preseason multiple biomechanics testing and dimension reduction for injury risk surveillance in elite female soccer athletes: short-communication,” *Sci. Med. Footb.*, vol. 7, no. 2, pp. 183–188, Apr. 2023, doi: 10.1080/24733938.2022.2075558.

[22] E. Barba, D. Casamichana, P. Figueiredo, F. Y. Nakamura, and J. Castellano, “The Use of Principal Component Analysis for Reduction in Sleep Quality and Quantity Data in Female Professional Soccer,” *Sensors*, vol. 25, no. 1, p. 148, Dec. 2024, doi: 10.3390/s25010148.

[23] S. Gaudet, M. Begon, and J. Tremblay, “Cluster analysis using physical performance and self-report measures to identify shoulder injury in overhead female athletes,” *J. Sci. Med. Sport*, vol. 22, no. 3, pp. 269–274, Mar. 2019, doi: 10.1016/j.jsams.2018.09.224.

[24] N. Bhawsar and S. K., “Discriminant Analysis of Sports Personalities for Predicting Better Performance,” *Int. J. Comput. Appl.*, vol. 180, no. 13, pp. 1–6, Jan. 2018, doi: 10.5120/ijca2018916247.

[25] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification And Regression Trees*, 1st ed. Routledge, 2017. doi: 10.1201/9781315139470.

[26] M. Kuhn and K. Johnson, *Applied Predictive Modeling*. New York, NY: Springer New York, 2013. doi: 10.1007/978-1-4614-6849-3.

[27] H. Abdi and L. J. Williams, “Principal component analysis,” *WIREs Comput. Stat.*, vol. 2, no. 4, pp. 433–459, Jul. 2010, doi: 10.1002/wics.101.

[28] MacQueen, J., *Some methods for classification and analysis of multivariate observations. Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability,* vol. 4. 1967. [Online]. Available: https://books.google.co.uk/books?hl=en&lr=&id=IC4Ku_7dBFUC&oi=fnd&pg=PA281&dq=MacQueen,+J.+(1967).+Some+methods+for+classification+and+analysis+of+multivariate+observations.+Proceedings+of+the+5th+Berkeley+Symposium+on+Mathematical+Statistics+and+Probability,+1,+281-297.+&ots=nQ_cJ-FfmP&sig=fLXQbNAbUzszuTMOM8L-UrPtg7Q&redir_esc=y#v=onepage&q&f=false

[29] R. A. Fisher, "THE USE OF MULTIPLE MEASUREMENTS IN TAXONOMIC PROBLEMS," *Ann. Eugen.*, vol. 7, no. 2, pp. 179–188, Sep. 1936, doi: 10.1111/j.1469-1809.1936.tb02137.x.

[30] Y. Michailidis *et al.*, "Influence of Playing Position on the Match Running Performance of Elite U19 Soccer Players in a 1-4-3-3 System," *Appl. Sci.*, vol. 15, no. 15, p. 8430, Jul. 2025, doi: 10.3390/app15158430.

[31] R. Akenhead and G. P. Nassis, "Training Load and Player Monitoring in High-Level Football: Current Practice and Perceptions," *Int. J. Sports Physiol. Perform.*, vol. 11, no. 5, pp. 587–593, Jul. 2016, doi: 10.1123/ijspp.2015-0331.

[32] M. Buchheit, "Monitoring training status with HR measures: do all roads lead to Rome?," *Front. Physiol.*, vol. 5, 2014, doi: 10.3389/fphys.2014.00073.

[33] P. E. Shrout and J. L. Fleiss, "Intraclass correlations: Uses in assessing rater reliability.," *Psychol. Bull.*, vol. 86, no. 2, pp. 420–428, 1979, doi: 10.1037/0033-2909.86.2.420.

[34] P. J. Rousseeuw, "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis," *J. Comput. Appl. Math.*, vol. 20, pp. 53–65, Nov. 1987, doi: 10.1016/0377-0427(87)90125-7.

[35] R. Tibshirani, G. Walther, and T. Hastie, "Estimating the Number of Clusters in a Data Set Via the Gap Statistic," *J. R. Stat. Soc. Ser. B Stat. Methodol.*, vol. 63, no. 2, pp. 411–423, Jul. 2001, doi: 10.1111/1467-9868.00293.

[36] Ron Kohavi, “A Study of Cross-Validation and Bootstrap for Accuracy Estimation and Model Selection,” *Int. Jt. Conf. Arti Cial Intell.*, 1995, [Online]. Available: https://www.researchgate.net/profile/Ron-Kohavi/publication/2352264_A_Study_of_Cross-Validation_and_Bootstrap_for_Accuracy_Estimation_and_Model_Selection/links/02e7e51bcc14c5e91c000000/A-Study-of-Cross-Validation-and-Bootstrap-for-Accuracy-Estimation-and-Model-Selection.pdf

[37] G. H. van der Maaten, “Visualizing Data using t-SNE,” 2008, [Online]. Available: https://www.jmlr.org/papers/volume9/vandermaaten08a/vandermaaten08a.pdf?fbcl

[38] L. Breiman, “Random Forests,” *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, Oct. 2001, doi: 10.1023/A:1010933404324.

APPENDIXES

Appendix A

Table A1: Skewness and Kurtosis of Performance Metrics

Skewness and Kurtosis of Performance Metrics

Metric	Skewness	Kurtosis
distance_m	-0.707	1.823
hir_m	1.643	7.247
sprint_m	1.562	2.789
accels	0.528	0.521
decels	0.762	0.970
session_load	-0.308	0.395
top_speed	-1.779	9.133
avg_speed	-0.658	1.534
duration_mins	1.568	3.345
intensity_ratio	1.675	8.579
work_rate	0.005	0.127

Table A2: Outlier Detection Summary (IQR Method)

Outlier Summary for Performance Metrics

Metric	N_Outliers	Percent
duration_mins	33	8.35
avg_speed	21	5.32
hir_m	18	4.56
distance_m	14	3.54
sprint_m	14	3.54
decels	10	2.53
intensity_ratio	9	2.28
session_load	7	1.77
accels	5	1.27
top_speed	5	1.27
work_rate	4	1.01

Appendix B

Table B1 (Full Match vs Training Results)

Fixed Effects – Match vs Training

Metric	Estimate (Match – Training)	p-value
hir_per_min	+1.522	<0.001
sprint_per_min	+0.087	0.384
accels_per_min	-0.088	<0.001
decels_per_min	+0.025	0.271
distance_per_min	+14.674	<0.001
intensity_ratio	+0.012	<0.001
sprint_ratio	-0.001	0.426
accel_decel_ratio	-0.212	<0.001
work_rate	+3.475	<0.001
speed_endurance	-0.853	<0.001

Table B2 (Full Season Month Results)

Metric	Estimate (per month)	95% CI	p-value
sprint_ratio	+0.0010	[0.0002, 0.0018]	0.011
speed_endurance	+0.1637	[0.054, 0.273]	0.003
accels_per_min	-0.0156	[-0.025, -0.006]	0.002
decels_per_min	-0.0134	[-0.026, -0.001]	0.039
hir_per_min	+0.044	[-0.080, 0.168]	0.480
sprint_per_min	+0.053	[-0.003, 0.109]	0.064
distance_per_min	-0.845	[-1.775, 0.085]	0.075
intensity_ratio	+0.001	[-0.001, 0.003]	0.284
accel_decel_ratio	-0.003	[-0.030, 0.024]	0.826
work_rate	-0.093	[-0.248, 0.062]	0.240

Appendix C

Table C1: PCA Component Scores for All Players

PCA Component Scores for All Players

Athlete	Position	PC1	PC2	PC3
Angus Millar	Striker	-6.475	0.033	1.067
Max	Striker	-1.960	-0.710	-0.988
Glenn Turner	Striker	-0.839	-4.616	-1.277
Luca Gardiner	Midfield	-0.944	-0.113	1.494
Ben Young	Striker	-0.156	-0.256	0.582
Alex MacGregor	Striker	0.343	-1.381	-0.698
Mathew Young	Striker	0.048	-0.373	-0.091
Stephen MacKenzie	Midfield	-0.746	1.955	0.616
Andrew Docherty	Midfield	1.364	-2.575	1.207
Angus Sinclair	Midfield	-1.186	0.638	0.137
Sean Baker	Back	-0.756	0.788	-1.391
Finn Halliday	Striker	2.764	0.212	1.046
Fergus fitzgerald	Striker	1.501	-0.190	0.806
Joe Russell	Midfield	-0.490	1.540	-0.914
Oscar lochrin	Striker	3.037	-1.403	0.668
Anup Valmiki	Midfield	1.492	0.899	-2.708
Jedd Campbell	Back	0.264	1.697	0.070
Louis gardiner	Midfield	0.779	2.548	-0.238
Calum Darroch	Midfield	1.960	1.306	0.613

Appendix D

Table D1: Players in Each Cluster

Player Archetypes Based on PCA Clustering

Archetype	Players
Specialists (n=6)	Glenn Turner, Alex MacGregor, Andrew Docherty, Finn Halliday, Fergus fitzgerald, Oscar lochrin
Balanced Performers (n=12)	Max, Luca Gardiner, Ben Young, Mathew Young, Stephen MacKenzie, Angus Sinclair, Sean Baker, Joe Russell, Anup Valmiki, Jedd Campbell, Louis gardiner, Calum Darroch
Elite Performer (n=1)	Angus Millar

Table D2: Misclassified Players – LDA

Misclassified Players Based on LDA Model

Athlete	Actual	Predicted	PC1	PC2	PC3
Luca Gardiner	Midfield	Striker	-0.944	-0.113	1.494
Andrew Docherty	Midfield	Striker	1.364	-2.575	1.207
Sean Baker	Back	Midfield	-0.756	0.788	-1.391
Finn Halliday	Striker	Midfield	2.764	0.212	1.046
Jedd Campbell	Back	Midfield	0.264	1.697	0.070